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Analysis and comparison of Photovoltaic (PV) Inverters for Grid Applications

Progress Report

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Statement of Authentication

This thesis contains no material that has been accepted for the award of any other degree or diploma and that, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person, expect when due reference is made in the text of this thesis.

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Abstract

The increasing need for renewable energy solutions arises as the concerns over greenhouse gas emissions, pollution and fossil fuel grow. A major contributor to climate change is the production of destructive energy sources like fossil fuels, transitioning towards cleaner energy approaches is imperative. One of the solutions could be solar energy as it has minimal impact on the environment and produces abundant energy. Conventional PV inverters use a transformer to create a galvanic isolation, but this reduces the efficiency, increase complexity and raises manufacturing cost. By removing the transformer, the drawback of the conventional approach is addressed, however, it also creates common-mode leakage current through the parasitic capacitance, which, if is this left unaddressed, increases power losses, electromagnetic interference, and safety risks.

This report dives deeply into the single phase transformerless photovoltaic (PV) inverter topologies for grid applications, concentrating on the configuration and modulation techniques used to suppress the leakage current and improve conversion efficiency. The conventional H-bridge topology was selected with both bipolar and unipolar Sinusoidal Pulse Width Modulation (SPWM), the baseline configuration as the more advanced topologies are heavily inspired by it.

Furthermore, the Virtual DC Bus topology was studied as the primary focus of this progress report, along with the H6 and 5-Level Mid-Point Clamped HERIC topologies for comparison. Focal points were common voltage behaviour, leakage current, power losses, efficiency and complexity. MATLAB/Simulink, an industry standard simulation software, was utilised to model and analyse the unipolar and double frequency SPWM techniques for the Virtual DC Bus topology, to assess output voltage quality, switching ripple, and filtering requirements.

Chapter 1

1.1 Introduction

This is an era in which concerns about non-renewable energy are increasing and the use of the electricity has increased dramatically, something that hasn't been seen throughout history. Inevitably, non-renewable energy will be depleted, so switching to different cleaner energy is imperative.

One of the most promising renewable energy sources is the Photovoltaic (PV) source, due to its clean, reliable and emission free nature. Energy from solar panels has become a norm in residential, commercial and utility applications. PV systems are effectively integrated into to the power grid and are one of the most researched fields in electrical engineering. One of the drivers of PV system is linked with the decreases in solar module cost, improvements made to the design, and increasing desire to minimise greenhouse gas emissions, which has resulted rapid growth of installed PV capacity since 2010.

The PV array generates direct current (DC) electricity, the voltage varies according to the solar irradiance, temperature and operating voltage. This DC power cannot be injected into the grid system because the grid uses alternating current (AC) power. Grid systems use AC power because it was difficult to change the voltage of a DC source, especially during the earlier days, and changing it now is not technically or financially feasible. Grid connected systems are essential for delivering energy directly to the grid without the need for battery storage. Eliminating batteries reduces cost, maintenance and since batteries have a short lifetime.

Solar energy is the leading renewable energy source in terms of capacity, in Europe the power output of PV generated energy was 13.3 GW, compared with wind power having 9.3 GW in 2010 according to (Meneses, D. *et al.* 2013), this is because sunlight is available everywhere, unlike wind and rivers. Wind turbines are only useful where there is strong wind and hydropower requires flowing water, where rivers must have an elevation drop. Hydropower and wind turbines depend on a specific location, but PV systems are highly adaptable to geographic location. PV system don't require massive deconstruction of buildings, making installation less complicated and they can be placed anywhere like residential home, commercial buildings and farmland. They require little time to be installed as they are made of prefabricated modules.

According to (Kouro, S. *et al.* 2015) the total cost of PV systems has reduced by 68% between 2000 to 2012. This reduction in cost will continue and PV system will become more efficient as the technology advances. One of the main reasons the world hasn't fully adopted PV systems is the cost, building solar energy system is expensive compared to the traditional methods, due to the manufacturing cost of the PV panels. However, PV technology has no moving parts, and its hardware is robust, has a long lifetime and requires very low maintenance. The cost after manufacturing is almost nothing.

The only real problem with for PV systems is that efficiency of a commercial PV panel is around 15- 20%, as stated by (Kerekes, T 2009). This efficiency is relatively low because the traditional transformer PV inverters provide galvanic isolation between the photovoltaic array and the electrical grid, this improves electrical separation and limits leakage current. However, Transformer and winding losses reduce efficiency and the energy delivered to the grid. Transformers are also heavy, resulting in lower power density and making transportation, installation, and maintenance more challenging.

By removing the transformer from the PV inverter, it reduces copper and magnetic field losses, which increases the efficiency. Without a transformer, manufacturing and installation are reduced. However, eliminating the transformer from the inverter creates common mode leakage current that flows through the parasitic capacitance, this occurs naturally and cannot be ignored. If left untreated, rapid changes in common mode voltage can drive unwanted current, increasing power losses and electromagnetic interference. To rectify this, various advanced topologies such as HERIC, H6 and Virtual DC Bus have been developed, this report will focus on the Virtual DC Bus topology. For simulation MATLAB/Simulink will be utilised for simulation, to model power electronic circuits.

1.2 Background

The PV array, DC link, inverter, output filter, measurement and control unit, and protection equipment are the major components of a grid connected PV system. PV arrays generate direct current (DC), which differs in power due to its reliance on irradiance, temperature, shading, and terminal voltage. The PV inverter converts DC to alternating current and the filter reduces switching harmonics, the power is then distributed to the grids.

A PV cell is a device that converts sunlight or artificial light into electrical energy. Sunlight is made up from photons, when solar panels are exposed to sunlight, these photons penetrate the cell and are absorbed by the semiconductor material. An electron is dislodged from

the semiconductor material when there is sufficient energy is absorbed, creating a movement of electrons. This movement of electrons toward the front surface of the PV cell creates an imbalanced charge difference between front and back surface, creating a voltage difference. When the cell is connected to a load, the electrons flow through the circuit, which producing electrical energy.

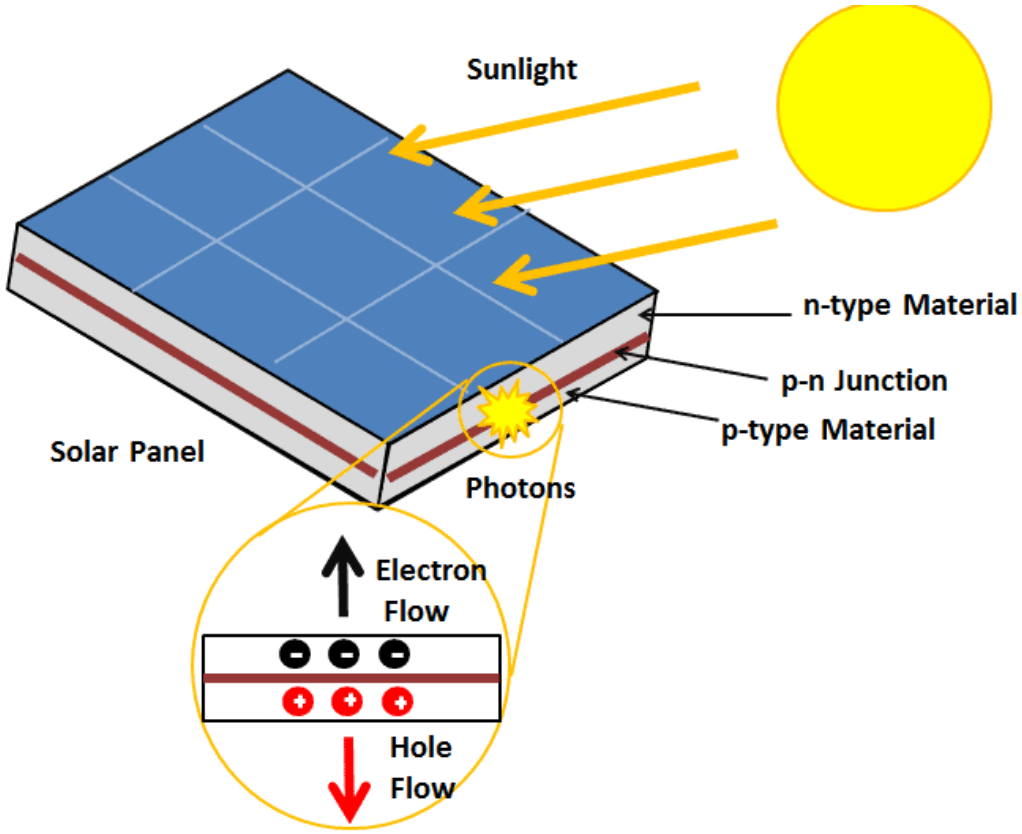


Figure 1 – (Donev et al., 2024); Photovoltaic Cell Operating Energy Conversion Process

A single photovoltaic cell generates around 1 to 2 watts, when multiple of these cells are connected to multiple of them, they increase the power output by a vast amount, this is called a PV panels. A PV array made of multiple of PV panels. Since the flow of electrons is in one direction, this results in DC current. DC current cannot be injected into the grid because it would waste energy. Both DC and AC lose power when transmitted, but AC is generally used in the grid due to its lower complexity and cost. Wires used in transmission lines have internal resistance even though the resistance is quite small. When transmitting a huge amount of electricity, the current can get quite high, using Ohms Law, (V = I * R), resistance in wire can consume significant power. By increasing the voltage and reducing the current, in theory, power losses during the transmission lines should be reduced. It was extremely difficult to

change the voltage level in DC, especially during 1800s, but its transformers can be used to step voltage up and down.

The Figure 2 below shows a comparation of the energy sources used between 1965 – 2025, with solar energy being the second highest rated energy source among renewable sources. The production of solar energy increased by approximately 30 times over the past decade. The graph shows that there was minimal growth for solar energy before 2010, followed by a rapid increase in solar energy generation between 2015 and 2025. The growth rate of solar energy is significantly greater than that of other renewable sources such as hydropower and bioenergy. Even though the hydropower remains the largest renewable electricity source; solar energy is closing the gap between them due to its strong growth rate. Moreover, the Figure 2 also highlights that solar energy has surpassed traditional energy sources such as the bioenergy and wind power. Therefore, the solar energy has the potential to become the dominant source of the electrical energy generation due to its rapid growth.

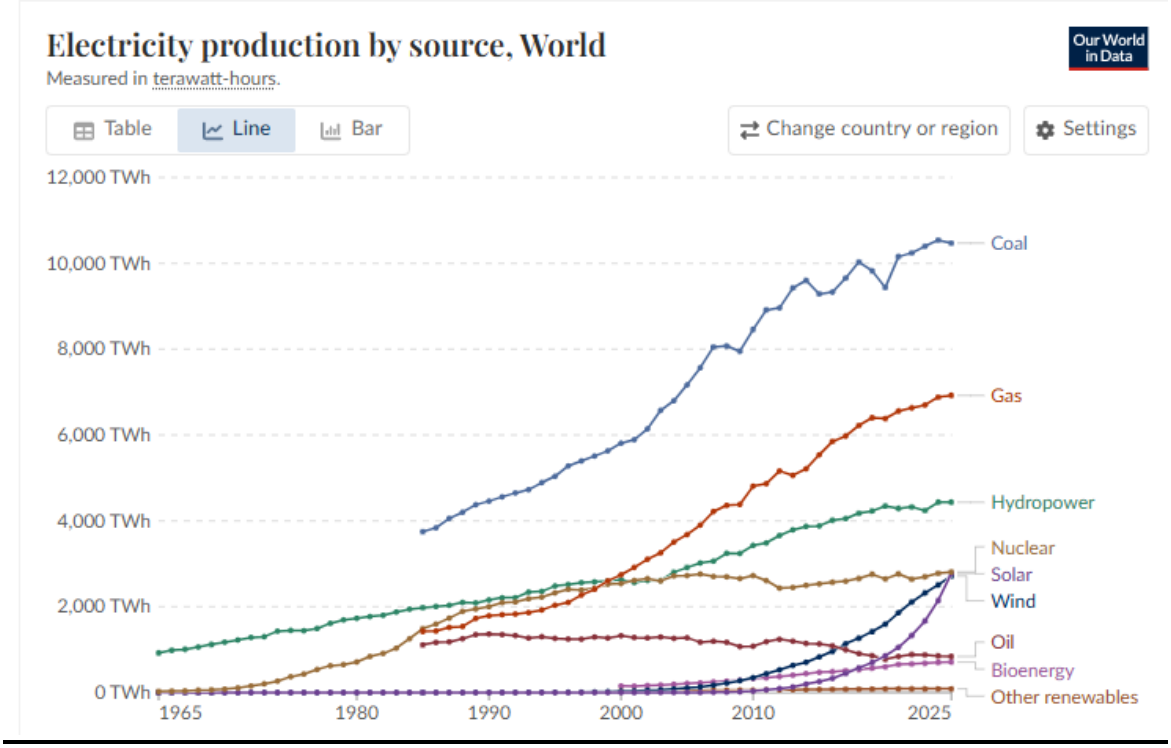


Figure 2 – Electricity production by source (2026); Type of Energy Sources Production in the World

Since the PV array can only produce DC current, and grids require AC current an inverter is required to convert DC to AC. An inverter uses switches, by rapidly turning them on

and off with SPWM causing the DC voltage to closely resemble a sine wave, The signal is then smoothed through filtering into AC power, which can be injected into the grid.

Smart grids are advanced distribution networks, unlike traditional grids which only supports one way energy flow, they support two way energy flow. Figure 3 shows smart grid communication flow, which allows it to distribute electricity from power plant, wind turbines, hydropower and solar panels. Residents can produce electricity for themselves and distribute energy back to the grid, since solar panels be installed in residential areas. Residents can potentially earn money or receive credit for their contributions. Smart grids can also detect faults, reduce energy loss, increase efficiency, and integrates well with PV systems, supporting sustainable and flexible electricity. The benefits of a smart grid over the conventional grid includes two-way communication, real time monitoring, fault detection, active response to demand, integrates with renewable energy, and better consumer engagement.

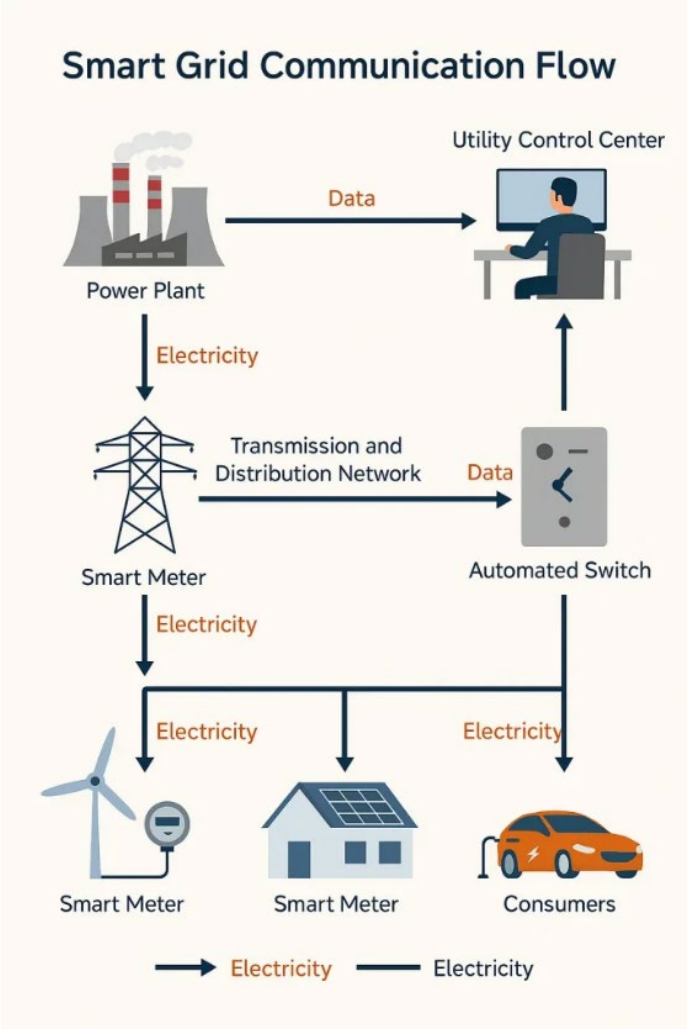


Figure 3 – IoTDunia (2025); Smart Grid Communication Flow

Chapter 2 Aims and Objectives

2.1 Aim

The aim of this progress report is to investigate the transformerless photovoltaic inverter topologies for grid applications and evaluate how photovoltaic system configuration and modulation technique can be utilised to suppress common mode leakage current without decreasing the conversion efficiency.

2.2 Objectives

1. To analyse the advantages and disadvantages of transformerless PV inverters topologies compared with traditional transformer-based system.

2. To evaluate the principles of the conventional H bridge inverter and its importances, as more advanced models are built upon it.

3. To simulate H bridge inverter topologies using Bipolar and Unipolar SPWM modulation techniques in MATLAB/Simulink, using the simulation results as baseline for this report.

4. To investigate how the absence of the transformer, along with the PWM switching frequency allows common mode leakage current to flow through the parasitic capacitance of the PV array and how it affects the performance and safety of the system.

5. To evaluate the flaw of unipolar and bipolar models, explaining why they are not suitable for industrial level PV inverters and why the advanced topologies are required.

Chapter 3 Literature review

3.1 Structure of PV System

APV panel is a DC power source that converts light energy into electricity. A PV panel consists of multiple of PV cells connected in series or parallel to achieve a greater power output. The PV cells are typically made from silicon semiconductor material. The conversion occurs when an electron is excited by a photon from the sunlight, this creates a flow of electrons which completes the circuit to produce the energy. The single diode equivalent circuit model contains an independent current source I_{ph} symbolising the photocurrent generated by sunlight, diode I_D represents the current released due to diffusion in the p-n junction. There are also internal losses, and leakage current is representing by a resistor R_{pv} to represent it.

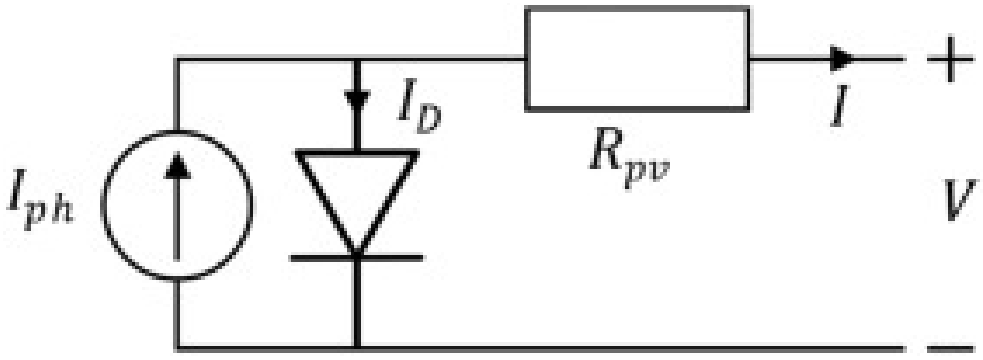


Figure 4 - Sopian, K.B. et al. (2004); Equivalent circuit diagram of a PV cell.

PV panels can be arranged in different configurations, the different combination of elements, panels and converters directly affects the cost, operation, complexity and efficiency of the PV system. The common PV configurations are according to (Kerekes, T 2009) are:

3.1.1 Type of inverters

Central inverter

The centralised configurations are used in the plants where the power requires more than 10kWp, PV panels are connected in series parallel configuration in huge numbers of them. Thyristor inverters were only used, after the IGBTs were introduced, they slowly replaced by thyristors for greater efficiency and cost savings. The common drawbacks are

- Need for high voltage DC cable

- Power loss due to Maximum Power Point Tracking (MPPT) and module mis-match
- Losses in the diodes and reduced due to relying on one inverter.

String inverters

String inverters are made of series connected solar panels and connected to separate inverters. They are paralleled and connected, there is no booster required if the string voltage is high enough, resulting in increased efficiency. The leading edges of string inverters compared with the central inverter are

- They required no diode, so there are no losses in the diode
- Each string has a separate MPPTs and therefore achieves a better yield
- Lower price due to economics of scale

Module inverter

A Module inverter is when each solar panels has a dedicated inverter and MPPT, resulting in maximised the power production. The superiority of this configuration is that there are no mismatch losses, it has finer optimisation than the string inverter in terms of the power extraction and the model is quite less complex as it has a plug and play trait. The disadvantage of this is that high voltage amplification may reduce the overall efficiency.

Multi String inverters

Multi String inverters are inspired by string and module inverters, they combine the advantages of both, having many DC-DC converters with dedicated MPPTs. The energy is then fed to a DC-AC inverter. This allows for one common grid connected inverter regardless of nominal data, size, technology, orientation. Inclination and weather condition of the PV system.

3.2 Parasitic capacitance of PV

Most of the PV panels are required to have a metallic frame to meet with the safety regulations and standards, which is done across almost all countries. This metallic frame and the surface area of the PV panels creates a parasitic capacitance as seen in Figure 5. The factors which influence the value the parasitic capacitance are surface area of the PV array, the distance between the PV cells and the module frame, and environmental factors that increase electrical conductivity such as dust, moisture, and humidity. According to (Kerekes, T 2009) the parasitic capacitance varies significantly under different conditions. Under optimal conditions the values

can be around in the picofarad range, however when the panel is exposed to the water, C_{pv} can increase by 60 times its original value. The parasitic capacitance of the PV arrays differs, with each kW of the installed PV panels ranging around 50 nF to 150 nF for typically silicon-based PV panel. Thin films module can reach up to 1µF/kW, this is caused by the metallic sheet which are embedded in them.

Every PV installation has parasitic capacitance; a safety issue arises when it leads to the leakage ground current and depends on the return path that exists in the circuit.

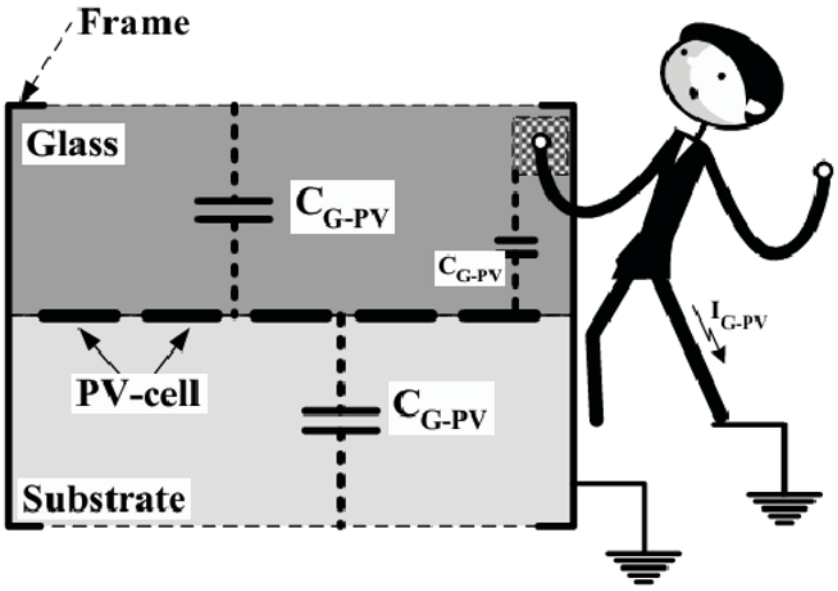


Figure 5 – Kerekes, T (2009), Parasitic capacitance in PV panels

3.3 Leakage ground current

Since there is an absence of galvanic isolation in transformerless topology between the PV array and grid. The fluctuation of the potential between the PV array and grid causes a direct path for the leakage ground current this is because PV panels are directly connected to the grid. The voltage fluctuations are caused by the parasitic capacitance when it charges and discharges, formed by between surface of the PV and grounded frame. Depending on the physical characteristics of the PV array and length of the DC cables can influence the resonance frequency of the circuit and form the resonant circuit, the DC lines that interconnects with PV array to the inverter, along with the parasitic capacitance.

When a person touches the surface of the PV array it can cause an electrical hazard. Depending on the inverter topology, the structure of the PV panels, and the modulation

techniques, a person may directly touches the PV panels without any safety precautions. A person could potentially complete the circuit as ground current could flows through the human body which may be result in a shock or a fatal injury if the current reaches a specific level.

Several common ways to minimise leakage current are

- Grounding the frame to reduce the capacitance
- Using a certain topology to reduce the leakage current significantly or eliminating leakage current and with a modulation strategy
- Disconnecting the inverter from the grid completely, to removes any leakage current usually done during service maintenance.
- Reducing the fluctuation in common mode voltage keeping it relatively constant can reduce the PV parasitic capacitance therefore the leakage current.
- Filtering the common mode can suppress the high frequency common mode components that causes the leakage current

3.4 Concerns of Transformerless PV System

The transformerless PV inverters are better option to due to the increased in efficiency, reduce the size and cost compared with traditional method. Nevertheless, the removal of the galvanic isolation solution may cause common mode ground leakage current through the parasitic capacitor, as illustrated in the Figure 6. The concerns are decreasing power conversion, increased in grid current distortion, degradation of the electromagnetic compatibility and safety threats are caused by the CM current. CM currents are formed by the IGBTs switches, filters, ground impedance Z_G and parasitic capacitance between the PV array and ground can are common source for the CM voltage. The CM voltage formula is derived from Figure 7.

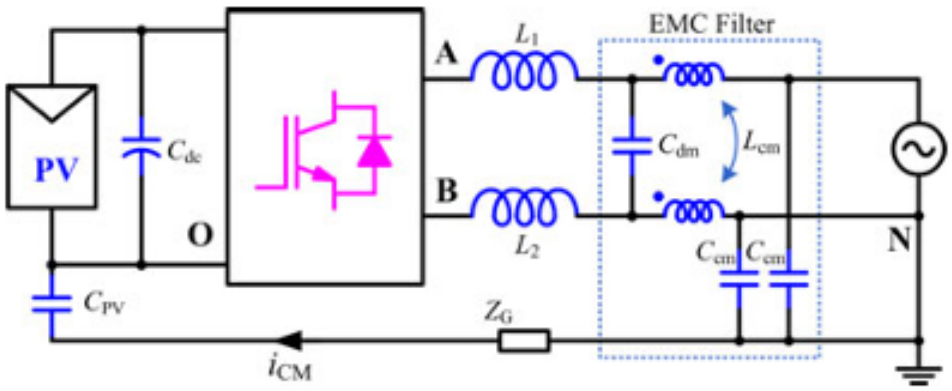


Figure 6 - Gu, Y. et al. (2013); CM current path for the transformerless PV inverter.

$$v_{CM} = \frac{v_{AO}+v_{BO}}{2} + (v_{AO} - v_{BO})\frac{L_2-L_2}{2(L_2+L_2)} (1)$$

Where v_{AO} and v_{BO} are the voltage potential difference of point A to O and point B to O, while the L_1 and L_2 acts as the output filters.

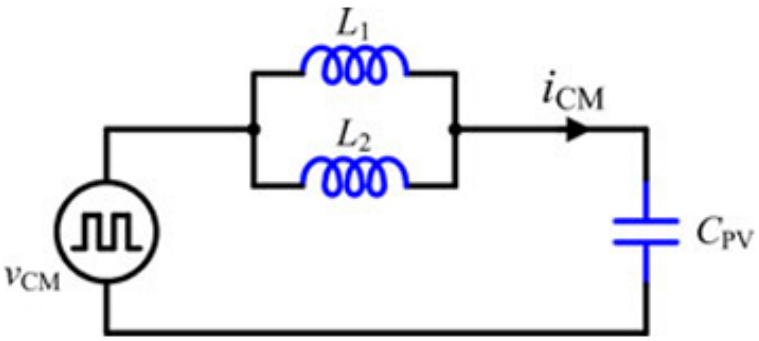


Figure 7 – Gu, Y. et al. (2013); Equivalent circuit for the CM current path.

3.5 Half bridge

The Half bridge inverter topology utilises two switches to connect upper or lower half of the DC link in phase with the grid, the DC link capacitors are permanently connected to the neutral conductor. The benefit of using the half bridge topology is that DC link is always connected to the neutral which improves the potential of the PV array and the voltage to the ground will be constant. A shortcoming of this topology is that DC link voltage is greater than the grid peak voltage by twice and the full DC link voltage is blocked by switches. In contrast to H bridge topologies, the volage is shared between two switches connected switches. Since T_1 is Operated at a high frequency in antiparallel with T_2, the converter output will be a bipolar voltage. This implies that converter efficiency will be decreased and requires a larger filtering component are required.

3.6 H Bridge Topology

The H bridge topology combines two half bridge together which results in single phase transformerless inverter. It commonly found in motors drives or uninterruptible power supplies (UPS). It is name by four switches with specific arrangement which resemble letter H. H bridge has four switches, usually MOSFETs or IGBTs, and labels S_1, S_2, S_3, and S_4. The switches work in diagonal pairs, when switch S_1 and S_4 are turned on, the current flows through the load in one direction, producing a positive voltage across the

load. In contrast, when switches S_2 and S_3 are turned on, the current flow in the opposite direction, producing negative voltage across the load. By alternating the switches states, the H bridge can be used to converts DC to AC. It is ineffective to manual change the state of the switches, but using PWM techniques to control it becomes much easier, the most common technique is bipolar and unipolar.

3.6.1 - H-Bridge topology with Bipolar PWM

The bipolar technique is easiest one to implement, it modulates switches T_1-T_4 and T_2-T_3 as its complementary, resulting the output of $+V_{DC}$ and $-V_{DC}$. For short period when inverter is not supplying voltage from the PV array, but current flows through from the stored magnetic energy from the grid inductor this is known as freewheeling interval. This reducing the conversion efficiency during this period the grid current finds a path and flows back to the DC link capacitor.

When the grid side filter inductor is on the line side this can be seen in Figure 8, high frequency components at switching frequency and can multiples of it, resulting in high amplitudes. This causes leakage current to be extremely high if the parasitic capacitance of the PV array is imposed by the high frequency voltage fluctuations. Therefore, this case with filter configuration is not ideal for transformerless PV systems.

When the grid side filter inductors are evenly distributed between the line and neutral conductors, the inverter produces a constant common mode voltage. The voltage to ground between the PV array will vary with grid frequency with an amplitude half of the peak value of the grid voltage. This reveals that transformerless PV system can use the H bridge with bipolar modulation, containing grid side filter with inductors evenly split between line and neutral, the only disadvantage is the decreases in conversion efficiency.

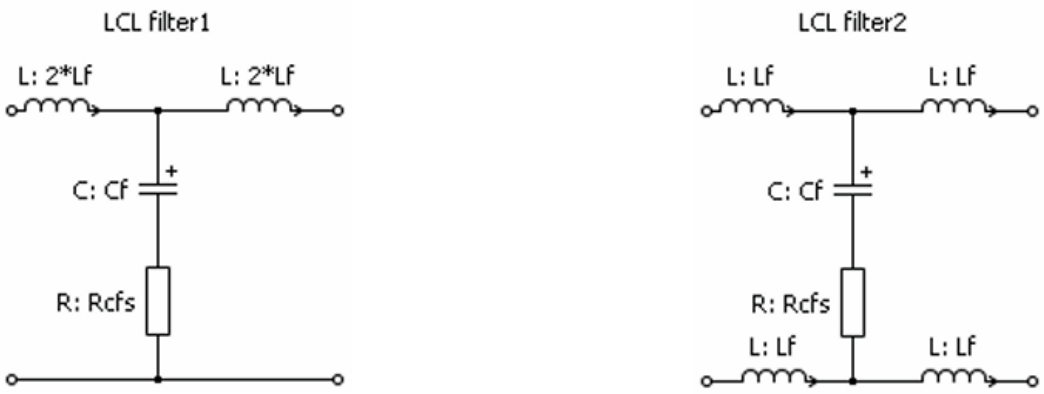


Figure 8 – Kerekes, T (2009) LCL filters

When the grid side filter inductors are equally distributed, in common mode voltage remains relatively constant because its balanced filter inductance in phase and conductors. One its shortcoming is the two-level output, has current ripples, large filter requirements and reduced in efficiency when returns to the DC link.

3.6.2 H-Bridge topology with Unipolar PWM

Unipolar PWM generates $+V_{DC}$, zero and $-V_{DC,}$ furnishing higher effective output frequency with three level output. This minimises the ripples in the current and allows smaller filter components. The benefit of this technique is that grid side filter element requires a smaller due three level unipolar waveform and increased effective switching frequency of the converter output voltage. During the freewheeling interval, the grid current flows through the short circuit output of the converter.

The limitation of unipolar of PWM is voltage to ground of the PV array and the ground leakage current. During the analysis of (Kerekes, T 2009) The leakage current peaks above 5A in the transformer less configuration, revealing enhanced differential mode waveform doesn't provide reasonable common mode performance. However, the numerical results are specific experimental system, as the leakage current is also affected by DC-link voltage, switching frequency, filter configuration, grounding and parasitic capacitance.

3.7 - Virtual DC Bus

The virtual dc bus concept contains two buses; each bus has two switches and capacitor. The real bus is directly connected to PV panels and labelled as node B, while the virtual bus is referred as node C. By connecting the grid neutral line to negative terminal of PV panel, the voltage is clamped to zero across the C_{PV}, prohibiting any leakage current. The state of the switch bridge can be either zero or $+V_{dc}$ regards to ground point N. The virtual DC bus purpose is to generate negative output voltage essential for inverter to operate. The voltage across the virtual bus can be regulated as the real bus, to develop proper transfer the energy between the two buses. As illustrated in Figure 9, voltage at midpoint C is either zero or $-V_{DC}$ as the positive terminal of the virtual bus is connected to the ground node N. The dotted line between the virtual bus and ground connection may be directly by a wire or indirectly using a power switch. When the node B and C are connected by the smart selecting switch, resulting three possible voltage levels of $+V_{DC}$, zero, and $-V_{DC}$ at point A. The structure of the

topology prohibits CM current, advanced modulation techniques like unipolar and double frequency SPWM can be utilised as there is no limitations on the modulation strategy.

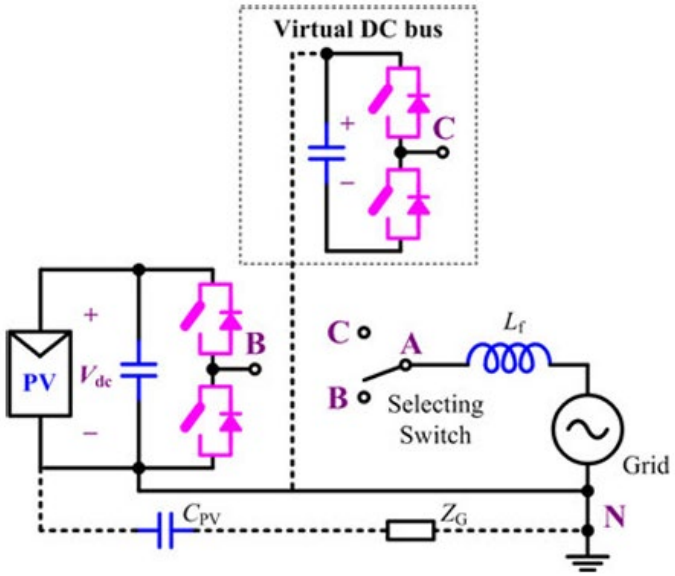


Figure 9 – Gu, Y. et al. (2013); Virtual DC Bus concept

The proposed topology has been derived from the virtual dc bus concept by Gu, Y. et al. (2013). The proposed methodology contains five IGBTS S_1-S_5 and one filter inductor L_f. The real dc bus is formed by PV panels and capacitor C_1 while capacitor C_2 provides the virtual bus. Using the switching capacitor technology the real dc bus charges the C_2 through S_1 and S_3 to supply a constant voltage. The advanced modulation technique, unipolar and double frequency SPWM are used to modulate the topology.

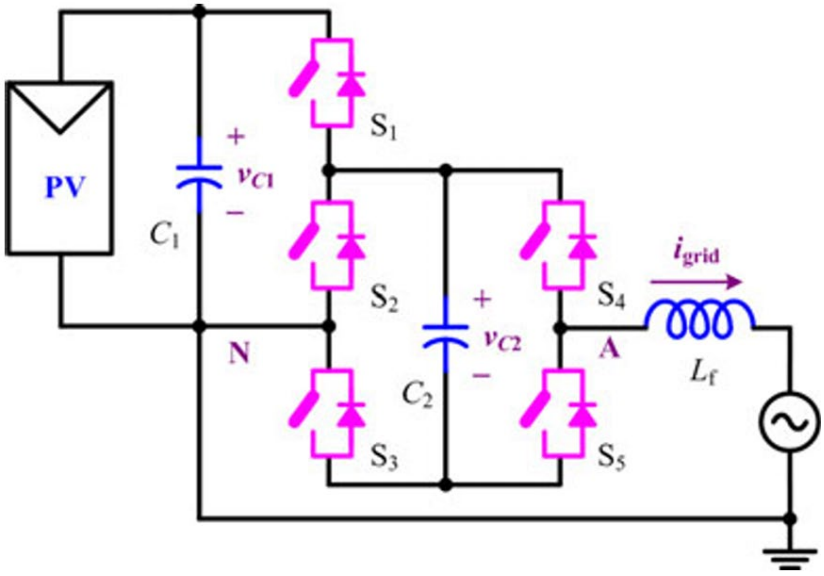


Figure 10 – Gu, Y. et al. (2013); Virtual DC Bus

3.7.1 Virtual DC Bus with Unipolar PWM

The proposed topology waveform for the unipolar SPWM is illustrated in Figure 11, the modulation wave ug and carrier wave uc are employed to generate the relative value for gate drive signal. When the grid cycle reaches the positive half, ug >0. S_1 and S_3 are turned ON and it's the complementary signal given to S_2, resulting S_2 to be turned OFF while S_4 and S_5 are being commutate complementally with carrier wave. The configurations of the C_1 and C_2 are parallel and the circuit changing its state between states 1 and 2 is shown in Table 1.

State 2 → State 1 → State 2 → State 1

When grid cycle is negative half, ug < 0. S_5 is turned on and S_4 is turned off. S_1 and S_3 are being synchronously commuted with carrier waves and complementary signal is given to S_2. The circuit changes its state between states 2 and 3. At state 3, the negative voltage is generated by virtual dc bus C_2 when S_1 and S_3 are turned OFF while the S_2 is turned ON. At state 2, C_2 is charged by DC bus through S_1 and S_3, this occurs when S_1 and S_3 are turned ON while the S_2 is turned OFF.

State 2 → State 3 → State 2 → State 3

Table 1 – Switching state for Virtual DC Bus

State	Switches ON	Switched off	Capacitor C2	Output Voltage
State 1	S1, S3, S4	S2, S5	Activated	$+V_{bus}$
State 2	S1, S3, S5	S2, S4	Activated	Zero
State 3	S2, S5	S_1, S3, S4	Activated	$-V_{bus}$
State 4	S2, S4	S_1, S3, S5	deactivated	Zero

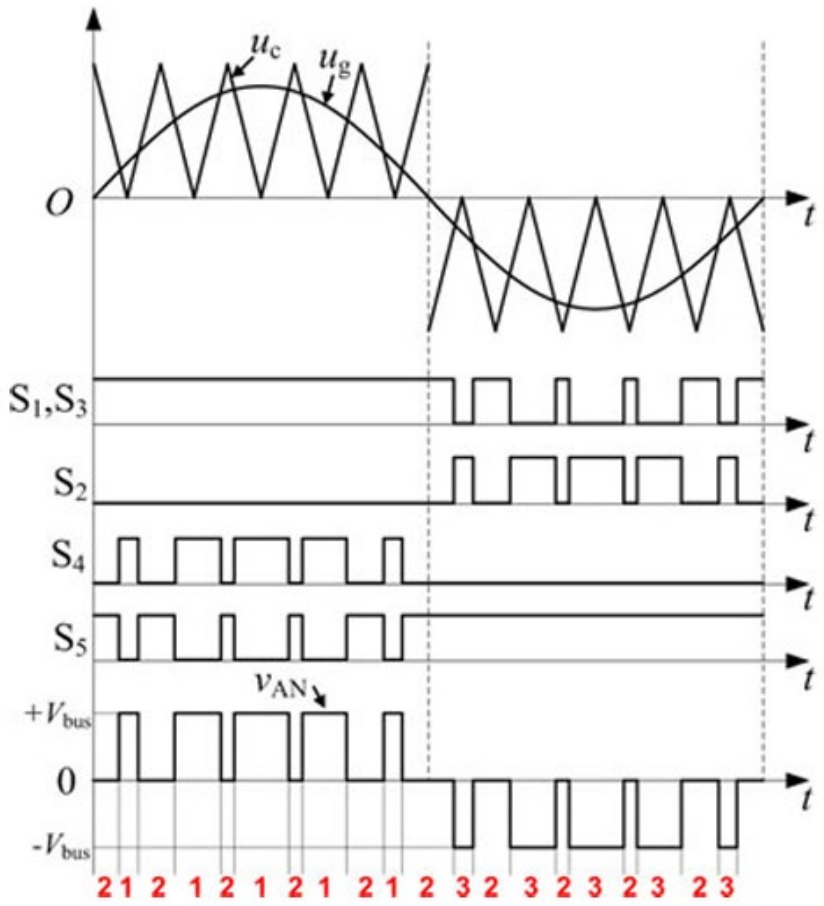


Figure 11 – Gu, Y. et al. (2013); Unipolar SPWM for Virtual DC Bus

3.7.2 Double Frequency SPWM

The proposed topology waveform for the Double Frequency SPWM is applied to achieve greater equivalent switching frequency as illustrated in Figure 12. The five power switches are divided into two parts and using two inverse sinusoidal waves to modulate. S_1, S_2 and S_3 are modulated with ug_1, while the rest are modulated with ug_2.

When the grid cycle reaches the positive half, the circuit changes its state between states 4, 1, 2 and 1 and the output voltage Van varies in pulse duration between +Vbus and zero with double of the carrier frequency.

State 4 → State 1 → State 2 → State 1

When the grid cycle reaches the negative half state the circuit changes between states 4, 3, 2 and 3 and the output voltage V_{an} varies in pulse duration between zero and $-V_{bus}$ with double of the carrier frequency.

State 4 → State 3 → State 2 → State 3

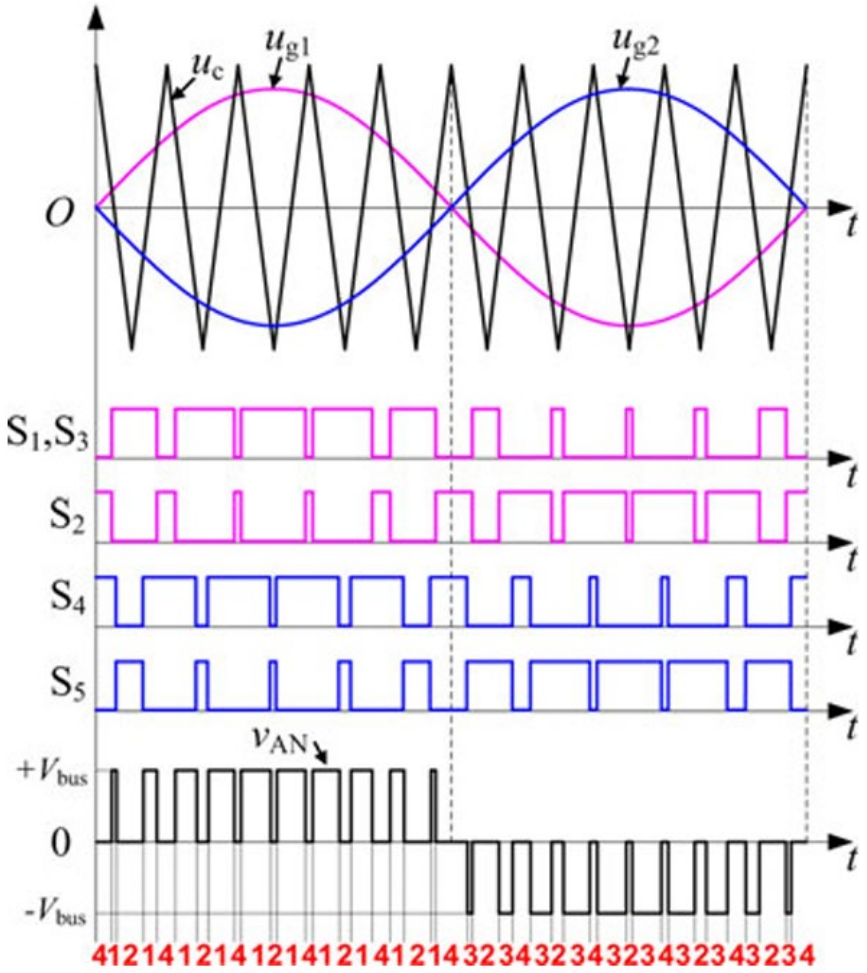


Figure 12 – Gu, Y. et al. (2013); Double frequency SPWM for the Virtual DC Bus

3.7.3 Comparison of both SPWM Virtual DC Bus

Both modulation strategies have their own strengths and limitations. The double frequency SPWM has higher equivalent switching Frequency, as result the size and weight of filter can be reduced. Whereas the unipolar SPWM can assure that real bus charging the virtual dc bus C_2 by every switching cycle, this allows the stress on the S_1 and S_3 to reduce that are caused by the operation of the switched capacitor.

The states of the operations have no limitation in terms of the direction of the current igrid output, bidirectional current flow can be achieved through power switches with antiparallel diode. The topology is also immune to transient over voltage,

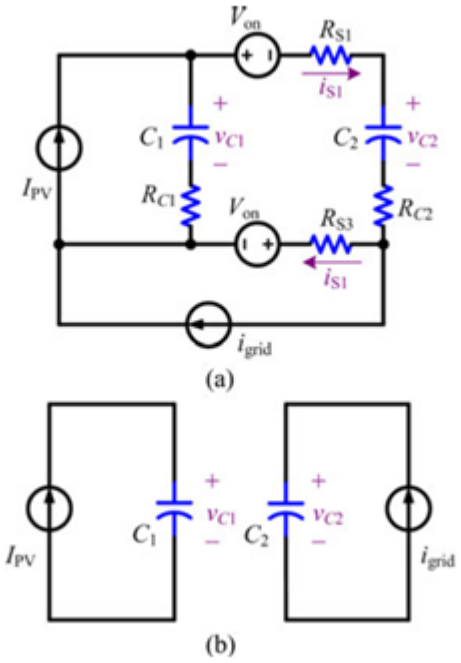


Figure 13 – Equivalent circuit for state 2 and 3: (a) state 2 (b) state 3.

Using the Figure 13 to derived

$$\text{State 2: } \frac{dv\Delta}{dt} = -\frac{v\Delta}{R_\Sigma C_S} - \frac{i_{grid}R_{S3}}{R_\Sigma C_S} + \frac{I_{PV}}{C_1} - \frac{I_{PV}R_{C1}}{R_\Sigma C_S} \quad (2)$$

$$\text{State 3: } \frac{dv\Delta}{dt} = \frac{I_{PV}}{C_1} - \frac{i_{grid}}{C_2} \quad (3)$$

3.7.4 Conduction Losses in the Power Devices

During the positive half cycle, the proposed inverter output current i_{grid} flows through S_1 and S_4 in state 1, i_{grid} flows through S_3 and S_5 at state 2, two switches are always on the path of i_{grid}. However, in the negative half cycle two switches are still on, but charging C_2 arises increases in current on S_1 and S_3, resulting in conduction losses. The proposed topology uses insulated gate bipolar transistors (IGBT) and diodes as power switches to reduce the conduction voltage drop $V_{on,}$ so the from the power switches losses can be treated as constant.

The conduction loss is directly proportional to the charge transferred through the power device during the grid cycle. The voltage of capacitor C_2 maintains balance in the grid cycle. The using switching cycle averaging method, to derive the extra charge.

$$Q_{ext} = \frac{I_m D_m T}{4} \quad (4)$$

Where I_m is the peak grid current, D_m is modulation index, and grid period is T. Since the S_1 and S_3 allows the Q_{ext} flows them, the losses can be shown by

$$P_{ext} = \frac{1}{2} V_{on} I_m D_m \ (5)$$

Therefore, the total conduction loss of the topology includes the conduction losses and losses by charging capacitor C2. The total conduction loss can be denoted as

$$P_{prop} = \frac{4}{\pi} V_{on} I_m + \frac{1}{2} V_{on} I_m D_m \ (6)$$

For the 220 V_{ac} system with 400 V DC bus, the modulation index D_m is 0.78, the resulting around 30% increase in conduction losses compared with conventional full bridge inverter. However, the real conduction is usually higher than the theoretical values because the stress current through the S_1 and S3 causes increase in the volage drop.

The switched capacitor contributions to the losses due to its equivalent series resistance (ESR). These losses are divided into two parts. The first part that contribute to the losses is the ripple current losses which is essentially to all voltage source inverter. Moreover, the second part is due to the inrush current caused by the switched capacitors during charging stage, these losses occur in the negative half cycle at state 2. The expected the ESR loss is approximately 1.6W at full load and ripple current loss is around 0.7W. To overcome these losses, a larger capacitance can be used and a small 10 uF film capacitor to eliminate the high frequency current component

3.7.5 CM current Elimination

The conventional full bridge based transformerless inverter topologies, aims to reduce the common mode (CM) current by disconnecting the AC side from the DC side during the zero-voltage level using extra switches. However, this disconnection not completely removes the parasitic capacitance on the power switches because of this high frequency CM current can still flow through the capacitors. These require extra filters to reduce the CM current.

The proposed topology eliminates the CM current from its virtual dc bus since the stray capacitance is bypassed directly between the PV panels and the ground.

Although, the conduction and switching losses that is comparable of proposed topology to the H5 circuit. The key disadvantage is the requirement that larger capacitors are required to reduce the ESR losses, by adding a thin film capacitor to deaden the high-frequency pulse current. The superiority advantage is that the elimination of common mode capability. Therefore, the proposed topology virtual dc bus concept furnishes about innovative concepts for developing transformerless PV inverter.

3.8 - H6 Inverter

The H6 transformerless inverter is an enhanced version of the H bridge topology, it is designed to overcome the shortcomings. The H6 is developed to reduce leakage current and conduction losses in PV systems. The H6 is high-performance inverter that effortlessly meets common mode voltage (V_{CM}) and differential mode (DM) requirements. The structure of the H6 consists of six switches and split DM inductors this can been seen the Figure 14. A H6 transformerless inverter topology is by complementing the bipolar PWM topology by employing two spilt inductors. A two supplementary switch and diode are placed together to form unidirectional cells. Then, unidirectional cells are engraved at the midpoint of the H4 inverter to create dedicated freewheeling path.

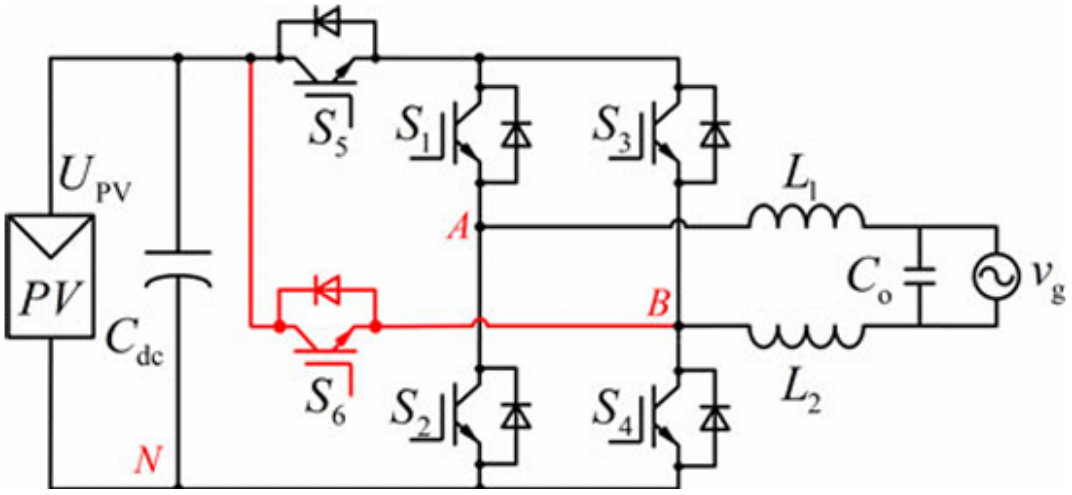


Figure 14 – Zhang, L. *et al.* (2014); Configuration of H6 topology

3.8.1 H6 with Unipolar PWM

In Mode 1, when the positive half period of the utility grid voltage is activated. S_1, S_4 and S_5 are turned ON while the remaining are turned OFF. The current flows through S_1, S_4 and S_5. (Formula Van = U_{pv}, $V_{bn} = 0$, $V_{ab} = V_{an} + V_{bn} = U_{pv}$) and common mode voltage is $V_{ab}/2 = 0.5U_{pv}$

In mode 2, occurs when the freewheeling mode in positive half period of the utility grid voltage. S_1 is turned ON, while the remaining switches are turned OFF. The inductor current freewheels through S_1 and the anti-parallel diode of S3. The terminal voltage is approximately equal to $V_{an} = V_{bn} = 0.5\ U_{pv}$, therefore the output is $V_{ab} = 0$ and the V_{CM} is approximately $0.5U_{pv}$.

During mode 3, when the negative half period of the utility grid voltage is activated. S_2, S_3 and S_6 are turned on, while remaining are turned off. Nevertheless, the inductor current

can only be through the S_2 and S_6. Even though the S_3 is that on state, current doesn't flow through it and as a result it does not produce any conduction losses in this state. The terminal voltages are as $V_{AN} = 0$, $V_{BN} = U_{pv}$ and $V_{AB} = -U_{pv}$ and V_{CM} is 0.5Upv.

Mode 4 is the freewheeling state that occurs during the negative half cycle of the utility gird voltage. All switches are turned off except S3, in this mode the inductor current freewheels flow through the S3 and antiparallel diode of S_1. The terminal voltage is approximately VAN = $V_{BN} = 0.5\ U_{pv}$, output voltage is around zero and $V_{CM} = 0.5U_{pv}$.

Table 2 – Switching State for H6

Mode/State	Grid Half Cycle	Condition	Switches ON	Switches OFF	VAB
1	Positive	Active	S_1, S_4, S_5	S2, S3, S6	+Upv
2	Positive	Freewheel-ing	S_1	S2, S3, S4, S5, S6	0
3	Negative	Active	S_2, S3, S6	S1, S4, S5	-Upv
4	Negative	Freewheel-ing	S3	S1, S2, S4, S5, S6	0

3.8.2 Power losses

The power losses are caused by switch loess, turning on and turning off switching losses, and other losses. The conduction loss is caused by power switches, or diode conducts current due to its internal resistance. Switching losses are from the semiconductor not having the instantaneously turned off and turn on capability, as a result the power dissipates as heat energy, so the higher switching frequency equal increases in switching loss. The other losses occur during the freewheeling, the current flows through the switches and antiparallel diode, creating diode conduction and reverse recovery losses. Since the energy is required to charge and discharge the switch gates during every switching cycle.

The power losses in the referenced H6 inverter are semi conductor and switching losses. As seen in Table 3, the calculated results for 1 kW power rating, the total power loss of th H6 topology is 25.37W compared with 27.71 for H5 and 23.03W for HERIC. The H6 topology reduces the conduction comparative to H5, it achieves it by the current flows through only S2

and S6 during active negative, even though is S3 is turn ON, this is why S3 has the lowest power loss.

Table 3 – Calculated Power Losses on Devices - Zhang, L. *et al.* (2014)

	S_1 (W)	S_2 (W)	S_3 (W)	S_4 (W)	S_5 (W)	S_6 (W)	Total losses(W)
H5	4.911	4.472	4.911	4.472	8.944	N.C.	27.71
HEIRC	4.472	4.472	4.472	4.472	2.571	2.571	23.03
H6	4.911	4.472	2.571	4.472	4.472	4.472	25.37

Note: U_{PV} = 380V.

3.8.3 CM Voltage and Leakage current.

The common mode voltage is produced while terminal A and B are switched among the positive and negative DC bus. The CM voltage can be noted as VCM = (VAN + VBN)/2.

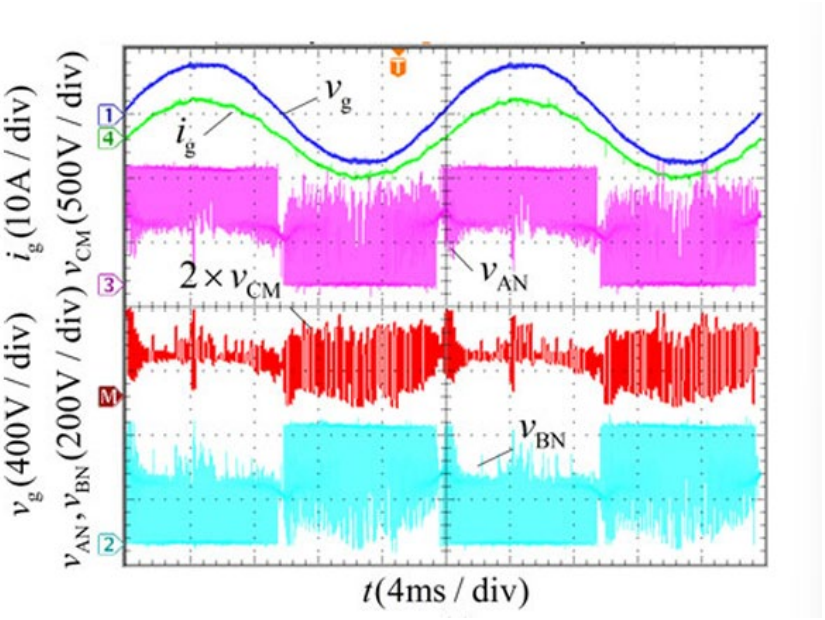


Figure 15 –Zhang, L. *et al.* (2014); CM voltage In H6 topology

As illustrated in Figure 15, the terminal voltage V_{AN} and V_{BN} are fluctuating along with active and freewheeling states. The $2V_{CM}$ waveform is relatively close to the DC link voltage, although the terminal voltage switches between different levels. The topology maintains near constant common mode voltage, this allows significant reduction in high frequency variation in V_{CM}, as a result very small amount of leakage current flows between the PV to ground.

Furthermore, during the zero-voltage freewheeling state, the inverter temporarily disconnects itself with the grid, to further restrict leakage current.

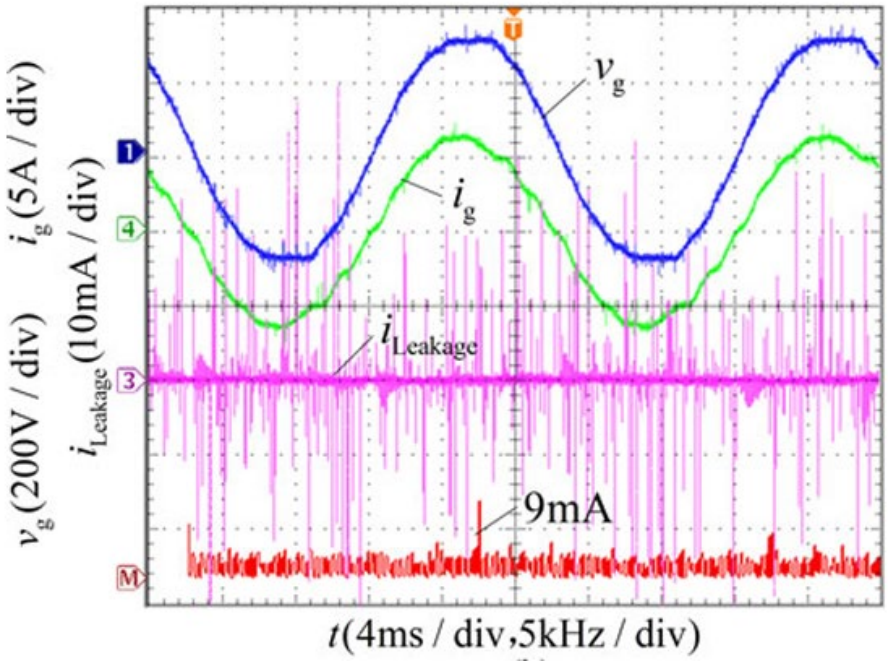


Figure 16 –Zhang, L. *et al.* (2014); Leakage current in H6 topology

Figure 16 shows small leakage current, but leakage current is still presented in the H6 inverter. In theory, the ideal common voltage should be near constant, but waveforms contain switching spikes and voltage oscillation, The Zhang, L. et al. (2014) states that this oscillation depends on the parasitic capacitance of the leakage current path and grid voltage. According to Ozkan, Z. and Hava, A.M. (2011) The leakage current for transformerless photovoltaic inverter is restricted below 300 mA for safety reason. H6 has significantly less leakage current with the standard value, the leakage current is approximately 9 mA at switching frequency as seen in the frequency domain waveform. Therefore, the H6 does not eliminate the leakage current, but reduce the fluctuation in V_{CM} and maintains V_{CM} around $V_{PV}/2$.

3.9 - 5-Level Mid-Point Clamped HERIC

Level 3 HERIC passive clamped topology was derived from the conventional H bridge by inserting AC side bypass. As shown in Figure 17, the topology consists of six power switches, four diodes, two DC link capacitors, two filter inductors and filter capacitor. The configuration of S1-S4 may look familiar, as it forms the H bridge and positive and negative output volage, while the remaining switching S5- S6 form the AC side bypass branch. The main purpose for

the AC bypass allows a path for the output current to flow through S5 and S6 during zero voltage states instead of the H bridge. Although the inverter reduces high frequency common mode voltage flucaiton, leakage current, switching losses and EMI, main limitation is that it produces larger voltage steps which causes high level of harmonic distortion.

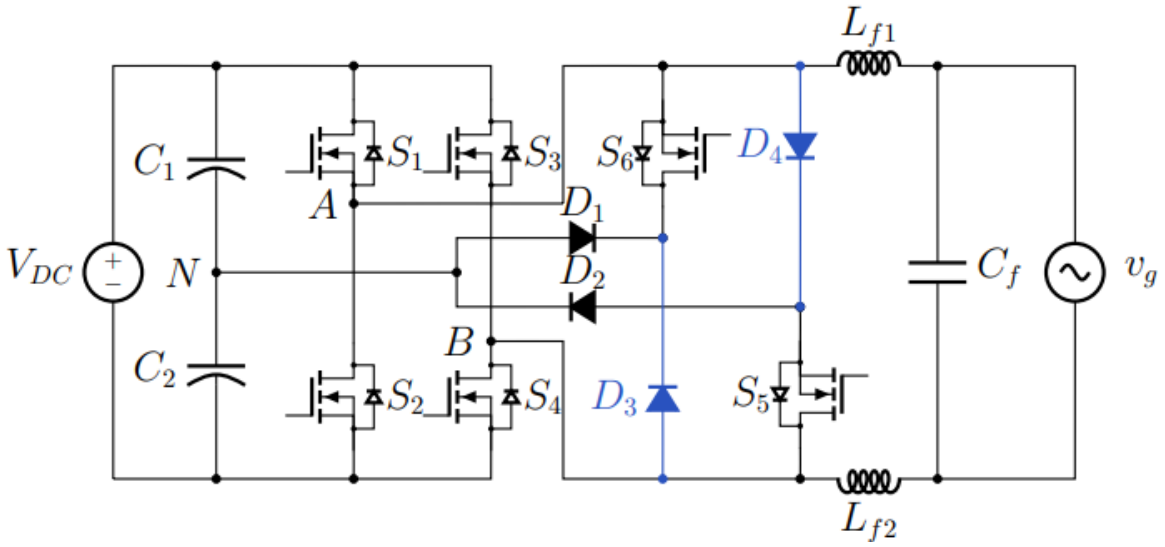


Figure 17 – Syasegov, Y.Y. *et al.* (2022); 3L HERIC passive clamped topology

Syasegov, Y.Y. et al. (2022) has modified the 3L HERIC passive clamped topology, where the D_3 and D4 are replaced with two new switches S6 and S7 to create 5-Level Mid-Point Clamped HERIC Inverter as seen in Figure 18. To maintain the output voltage of the inverter, the DC link capacitor voltage must be regulated at half of the DC link voltage. The switching states can be controlled by either level Shifted PWM or Phase Shifted PWM with two carrier signals.

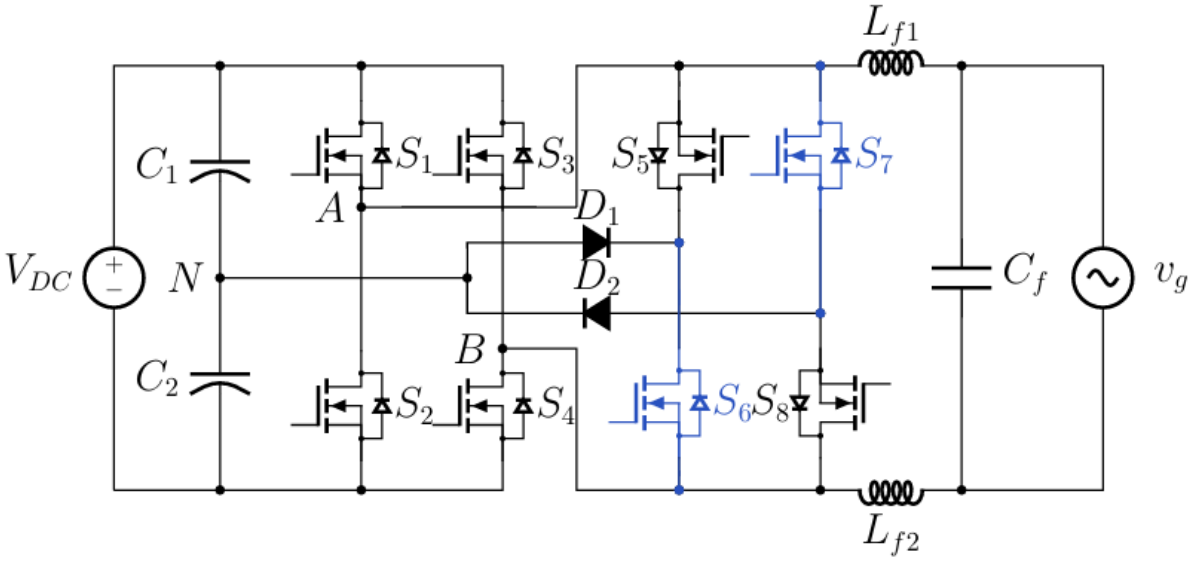


Figure 18 – Syasegov, Y.Y. *et al.* (2022); 5L HERIC topology

3.9.1 5L HERIC topology with Phase shift PWM

Phase shift PWM is preferred over Level-Shifted PWM due to improved performance. The switching pulses are produced by the two 180^0 phase shifted carrier waveform with sinusoidal waveform. The frequency of the output signal is twice the switching frequency with Phase shift PWM is utilized, as improves the frequency component of the load voltage and current.

The state of the operations are State A, B, C, D, E, F and G. In State A, S1 and S8 are ON and State B, S4 and S5 produces the same output voltage: $+V_{DC}/2$. These two states generate different current paths, to stabilize voltage across the DC link capacitors. In State C, S3 and S7 are ON and the remaining are turned off. State D, S2 and S6 are turned ON and rest of the power switches are turned off. State C and D produce - $V_{DC}/2$ and aid to balance the capacitor voltage. In State E, S1 and S4 are active and other switches are inactive to produce $+V_{DC}$, this is where the positive DC link voltage is applied. State F, S2 and S3 are ON and other power switches are OFF, providing $-V_{DC}$, this is where the negative DC link voltage is achieved. In State G, S5, S6, S7 and S8 are turned ON and remaining are turned off, this results in zero output voltage and decouples the DC source from the AC side. Therefore, all the operation state produces five different voltage levels: $+V_{DC}$, $+V_{DC}/2$, 0, $-V_{DC}/2$ and $-V_{DC}$.

Table 4 – Syasegov, Y.Y. *et al.* (2022); Switching States of the 5L HERIC Inverter

State	V_{out}	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8
A	$+V_{DC}/2$	1	0	0	0	0	0	0	1
B	$+V_{DC}/2$	0	0	0	1	1	0	0	0
C	$-V_{DC}/2$	0	0	1	0	0	0	1	0
D	$-V_{DC}/2$	0	1	0	0	0	1	0	0
E	$+V_{DC}$	1	0	0	1	0	0	0	0
F	$-V_{DC}$	0	1	1	0	0	0	0	0
G	0	0	0	0	0	1	1	1	1

The Phase shifted PWM is separates the output voltage cycle into four crucial operation areas. The state letters highlight how the inverter changes the voltage level to create a sinusoidal waveform.

Operation Area I occurs during the lower of the positive half cycle when the V_{out} is between 0 to $+V_{DC}/2$, the switching cycle is A-G-B-G. State A and B both produces $+V_{DC}/2$ while State G generates zero.

Operation Area II occurs during upper of the positive half cycle when the V_{out} is between $+V_{DC}$ / 2 to $+V_{DC}$, the switching cycle is E-A-E-B. State A and B both produce $+V_{DC}/2$ while State E generates $+V_{DC}$.

Operation Area III occurs during the lower half of the negative half cycle when the V_{out} is between 0 to - V_{DC} / 2, the switching cycle is C-G-D-G. State C and D both produce -V_{DC} / 2 while State G generates zero.

Operation Area IV occurs during the upper half of the negative half cycle when the V_{out} is between -V_{DC} / 2 to -V_{DC}, the switching cycle is F-C-F-D. State C and D both produce -V_{DC} / 2 while State F generates -V_{DC}.

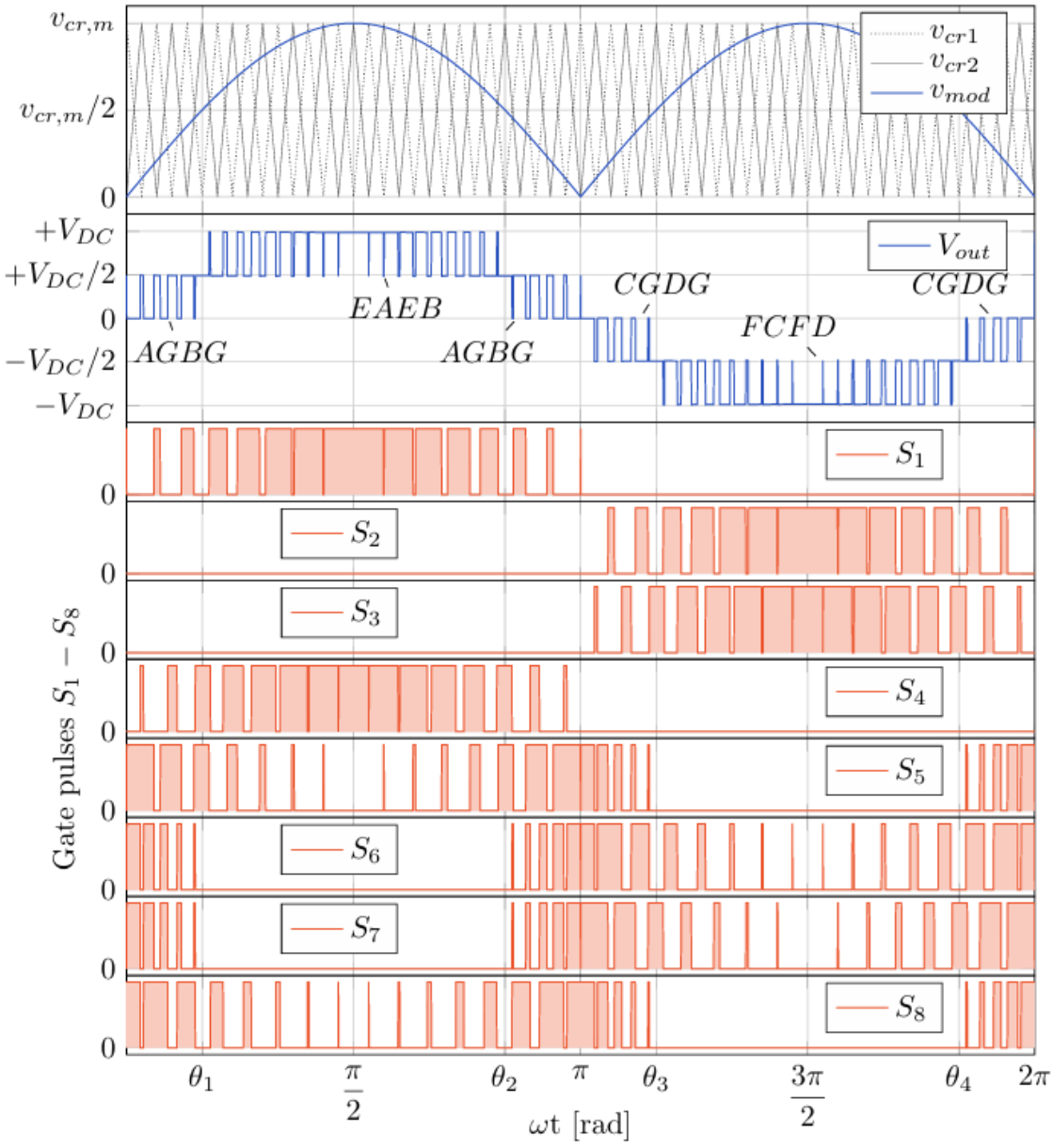


Figure 19 - Syasegov, Y.Y. *et al.* (2022); Phase-shifted PWM of the 5L HERIC Inverter.

3.9.2 Power losses and Efficiency

The power losses of the five-level HERIC inverter caused by the conduction loss, switching losses, diode losses and output filter inductor losses. The conduction loss is caused by MOSFET, or diode that conducts current due to its internal resistance. Switching losses are from the semiconductor not having the instantaneously turned off and turn on capability, as a result the power dissipates as heat energy, so the higher switching frequency equals increase in switching loss. Figure 20 shows the conduction and switching losses by semiconductors S1-S8, D1-D2, the devices operate at different frequencies and supplies current for different intervals.

The five-level waveform provides smaller voltage steps to the inductance filters. The phase shifted PWM, highlights effective output switching frequency of 2 f_{sw} this reduces the ripple current in inductor. Smaller filters are required since there is lower ripple and reduction losses from the filter. According to Syasegov, Y.Y. et al. (2022) the calculated efficiency includes semiconductor and filter inductor with equivalent series resistance (ESR). The efficiency of the HERIC increasing as the output power increasing, until it reaches 98.34 % at the rated power of 1.8 kW, the simulated injected current Total Harmonic Distortion (THD) was 0.32%. Therefore, the 5-Level Mid-Point Clamped HERIC achieves high efficiency while maintaining high quality voltage and current.

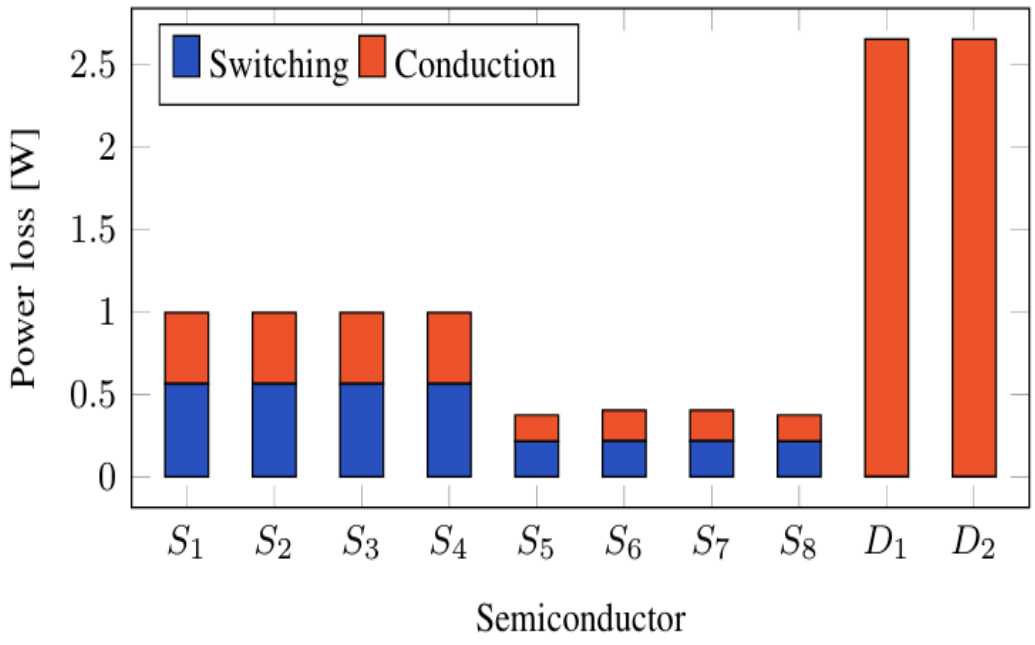


Figure 20 - Syasegov, Y.Y. *et al.* (2022); Loss distribution among semiconductors

3.10 Comparison of topologies

Table 5 - Comparison of topologies

Comparison	**Virtual DC Bus Gu, Y. et al. (2013);**	**H6 Zhang, L. et al. (2014)**	**Five Level HERIC Syasegov, Y.Y. *et al.* (2022)**
Voltage Levels	Five level: $+V_{DC}$, $+V_{DC}/2$, 0, $-V_{DC}/2$ and $-V_{DC}$	Three levels: $+V_{DC}$, 0 and $-V_{DC}$	Three levels: $+V_{DC}$, 0 and $-V_{DC}$
Modulation Strategy	Unipolar and double frequency SPWM	Unipolar SPWM	Phase shifted PWM or Level shifted PWM
Leakage current	0 mA	9 mA	N/A
Peak efficiency	Approximately 96%	97.09 %	98.34 %
Filter requirement	Requires one filter and one capacitor	Requires two output filter inductor and one capacitor	Requires two spilt inductors and one filter capacitor.
Key Advantage	Eliminate leakage current.	Provides near constant common mode voltage.	High efficiency, low THD and low current ripple.
Key Disadvantage	Requires large capacitors and charging current stress.	Doesn't completely remove leakage current.	Requires eight power switches, complex PWM and DC link.

Chapter 4 Research Question

1. **What are the advantage and disadvantage of transformerless topology over the topology which use transformer in PV system**

   Transformerless PV inverters provide high efficiency, lower cost, smaller size and weight much less, however, they are missing galvanic isolation, as a result it produces leakage current and safety hazards.

2. **Does modulation technique matter in providing output performance for PV inverter**

   The modulation technique affects output voltage quality, THD, efficiency, switching loss and leakage current.

3. **How does common mode voltage and parasitic capacitance create leakage current.**

   The PWM causes common mode voltage to fluctuate between PV and ground, this creates parasitic capacitance between PV and ground, producing leakage current.

4. **Why is leakage current being critical design challenge for transformerless H bridge PV inverters? How do Virtual DC Bus, H6 and HERIC topologies overcome this issue?**

   The leakage current is critical design challenge as the it increases power losses, electromagnetic interference (EMI) and safety risks. Virtual DC Bus, H6 and HERIC topologies reduce common mode voltage fluctuation, freewheeling path and virtual bus, significantly suppress or eliminate leakage current

5. **Why are transformer-based inverter solution used when transformerless PV inverters have higher efficiency and lower cost compared to transformer-based inverters.**

   Transformer based inverters provide galvanic isolation, safety and elimination of leakage current. Also, it aids in reducing switching noise and improve output waveform, application with strict safety regulation and dangers environments.

This study will use further research, literature review, analysis and simulation to compare the advanced topologies for single phase grid connected PV system. Also, it will evaluate their operating principle, advantages, limitations and their attempt to reduce leakage current. PWM techniques will be examined on efficiency, switching losses, output power quality and common mode voltage. Additionally, the study will also analysis the effects of parasitic capacitance and PWM switching frequency on leakage current and safety. Finally, comparison between

transformerless and transformer-based inverter will be in terms of cost, efficiency, size, galvanic isolation and safety.

Chapter 5 Methodology

5.1 Research Methodology

This report acquires an iterative research and simulation based to explore the transformerless photovoltaic invert topologies. The research was conducted with limited experience in writing a university engineering report, even though report writing was undertaken during high school, this research paper requires more critical and technical knowledge. Research at high school, information was led to single concrete answer, during this progress report the examined research papers had different circuit configurations, switching strategies, control methods, simulation parameters, and performance. Therefore, the progress report had be critically analysed rather than simply gathering information and reproducing it.

Using online tutorials and demonstration to understand the basic operation H bridge inverter and design of their configuration in Simulink. Each tutorial had different method to design the unipolar and bipolar models, the difference was arrangement of the switches, they used difference type of switches like IGBT and MOSFET. There was variation in the selection of the carrier waveform, since MATLAB doesn't have triangle waveform, this was replaced by repeating sequence or sawtooth waveform, the placement of the LC filter is also varied. Therefore, the online tutorials were used to get overview of the topology and implement the inverter.

The initial research was done through regular meetings with supervisor, which were twice a week. Meetings for project direction, progress, verifying the simulation design, and check the quality of the researched material. The supervisor gave overview of transformer, and transformerless inverter topologies used the PV system, after a discussion transformerless topologies were better choice due the simplicity and efficiency of the inverter. When looking for the reliably research paper, it was limit to the paper which were published in last 15 years and IEEE Xplore was primary used to find journal articles and conference papers.

Transformerless PV inverter. They used different circuit structures, semiconductor devices, switching frequencies, modulation techniques, filters, control systems, and components. To manage these variations, the conventional H bridge was selected as baseline topology since it was the foundation topology of the other configurations. Bipolar and unipolar SPWM were simulated using Simulink because they use the same four switch H bridged while being modulated.

The H bridge inverter is simple, inexpensive and using four switches, but the common mode voltage introduces parasitic capacitance which drives leakage current. This leakage current can't be ignored because it contributes to EMI, power losses and quality of the grid current.

Further research was conducted to investigate advanced transformerless topologies that can control the common voltage and reduce or eliminate the leakage current. During the research process, H6 was chosen first, subsequently Virtual DC Bus. A Five-Level Common-Ground Inverter with Reduced Switch Count was considered as third topology. However, after discussion with the supervisor, it was not recommended to be considered as third topology due the complexity of topology and it considered beyond scope for this progress report. Since there was already limited knowledge on working research paper, choosing that topology would have resulted obscurity. Therefore, it replaced by the HERIC topology

As result Virtual DC bus, H6 and HERIC are investigated on this progress report. These advanced topologies are directly built upon the conventional H bridge inverter to control or eliminate the path for common mode leakage current. All three of these advanced topologies are simulated using MATLAB/SIMULINK, each of them requires different modulation technique such as unipolar, double frequency and phase shift SPWM.

5.2 Simulation Methodology

5.2.1 Conventional H Bridge Inverter

This report dives deeply into the transformerless Photovoltaic (PV) Inverters for Grid Applications through a literature study analysis and evaluation of existing inverter topology. Using MATLAB/SIMULINK as a simulation tool due to its user friendly and extensive power electronics libraires, which allows to create accurate modelling and performance analysis.

Bipolar H Bridge inverter topology was developed because it is fundamental configuration which inspires numerous transformerless inverter topologies. The Bipolar H bridge inverter is developed using four IGBT switches arranged in H bridge configuration. A DC voltage was used to mimic the solar PV system. The SPWM signal was generated by the sine wave frequency of 50Hz and compared with high frequency sawtooth carrier waveform to produce a pulse to inverter operation. Since Simulink library doesn't have triangular waveform, it was replaced by sawtooth generator. The output of the relational operator was connected to the diagonal pair of IGBTs, while the complementary signal created by NOT gate and output of it is connected to the opposite pair, to ensure work properly switches of the H bridge and halt

short circuit. This configuration allows the two diagonal pair to work alternately, making the output to switch between $+V_{DC}$ and $-V_{DC}$.

LC filter consists of inductor, and capacitor is used to attenuate high frequency switching harmonics produce by the PWM steps. The series inductor stabilizes current flow and reduces current ripple; capacitor is connected in parallel to the resistive load to filter high frequency voltage. Therefore, the filter smooths the inverter output, only allowing 50 Hz component to pass to the load.

A resistor is placed at the end of the inverter to represent the load. Voltage measurement and scope are added to design to observe the output voltage. The inverter performance was evaluated by observing the filtered waveform, output voltage, switching ripple, harmonic distortion. Fourier transform analysis can be conducted to determine the harmonic content of the output voltage and effectives LC filter.

Table 6 - Values of Components used in simulation for H Bridge Inverter

Components	Values
DC Input Voltage (V_{dc})	100 V
Sinusoidal Reference Frequency (f_r)	50 Hz
Sawtooth Carrier Frequency (f_c)	20,000 Hz
Filter Inductance (L_f)	200 mH
Filter Capacitance (C_f)	20 µF
Load Resistance (R_{Load})	200 Ω
Load Inductance (L_{Load}):	0 H

Most of the topology are based on bipolar and unipolar H bridge topology, it those two inverters as baseline for the evaluating researched topologies which are discussed in the literature review. The value of the DC input voltage, sinusoidal reference frequency, sawtooth carrier frequency, filter inductance, filter capacitance, load resistance and load inductance are shown below for both unipolar and bipolar

5.2.2 Virtual DC Bus

The investigate Virtual DC Bus inverter in literature review was implemented in MATLAB/Simulink. The model consists of five IGBTs, S_1 to S_5, capacitors C_o and C_2 and inductors L_f and L_r. The real DC bus was formed using the DC source, while C_2 forms the virtual DC bus. C_2 is charged by the real DC bus through S_1 and S_3, allowing to create output voltage.

The circuit and modulation technique were reproduced using the prototype topology and SPWM waveform proposed by Gu, Y. et al. (2013) The C_1 was removed by simulated circuit mimic the PV model and since DC source is constant, doesn't vary in amplitude the C_1 can be removed because it is essential useless and doesn't contribute to any factor. Two modulation technique were implemented, unipolar and double frequency SPWM. The unipolar SPWM was more difficult to create because different switching state required, during the positive half cycle S_1 and S_3 were turned on and during the negative half cycle, the sinusoidal was compared with high frequency repeating sequence to the generate the PWM signal. However, the double frequency was relatively straightforward to implement, since it only required two sinusoidal waveforms.

Sawtooth waveform was initially used for replacement for triangular waveform carrier. When the simulation was shown to the supervisor, it was revealed that Simulink can't inherently produce triangular waveform, but there was way to create it. So, further research was conduct about creating triangular waveform. After reviewing online tutorial, triangular carrier using repeating sequence was created and compared with sinusoidal waveform to produce the required SPWM for both modulation techniques.

The several issues were met during the simulation. One of major issue the Simulink Refine Factor was accidentally set to 100,000, causing each simulation to take approximately 45 minutes when simulation was tried to stop, MATLAB would crash. This issue was unaware, several hours were spend on the reconfiguring the circuit, until MATLAB/Simulink help was utilised, it turns out the quality of the signal was extremely high. By adjusting refine factor in Model Settings to its default value.

Due to the refine factor, several changes were made that result in other issue was with output voltage ranging from + 740 V to – 400 V is was caused by improper circuit connections and improper filter values. According to Gu, Y. et al. (2013); highlights that grid voltage should be 220Vac in RMS, by converting to V_{peak} output was +311 V to – 311 V. After meeting with

supervisor, the topology was reconstructed, and the values and placement components were checked against referenced paper. Once issues in the simulation were fixed, the simulation produced the expected waveform.

The final model was verified and validated by the supervisor by examining the gate signal, inverter output voltage and current and voltage across the C2. After resolving the Refine Factor and circuit configuration issues, both unipolar and double SPWM performed accurately and produced desired output voltage.

Table 7 - Values of Components used in simulation for Virtual DC Bus

Components	**Values**
DC Input Voltage (V_{dc})	400 V
Sinusoidal Reference Frequency (f_r)	50 Hz
Switching Frequency (f_c)	20,000 Hz
Switched Capacitor (C_2)	940 µF
Filter Inductance (L_f)	8 mH
Filter Capacitance (C_o)	0.47 µF
Filter Inductance (L_r)	0.8 mH
Load Resistance (R_{Load})	10 Ω

Chapter 6 Results and discussion

The unipolar and double frequency SPWM Virtual DC Bus were simulated in MATLAB/Simulink to analyse the modulation technique how they convert DC power to AC power. Both models of Virtual DC Bus were identical in terms of IGBT switches, SPWM, LC filter, DC power supply and resistive load.

6.1 Unipolar Virtual DC Bus

Figure 21 show the simulated switching states and output voltage before the filter was used of the unipolar SPWM Virtual DC Bus inverter developed in MATLAB/Simulink. The modulation technique and circuit configuration is based on Gu, Y. *et al.* (2013), where the sinusoidal signal is compared with the high frequency triangular waveform to the PWM signals.

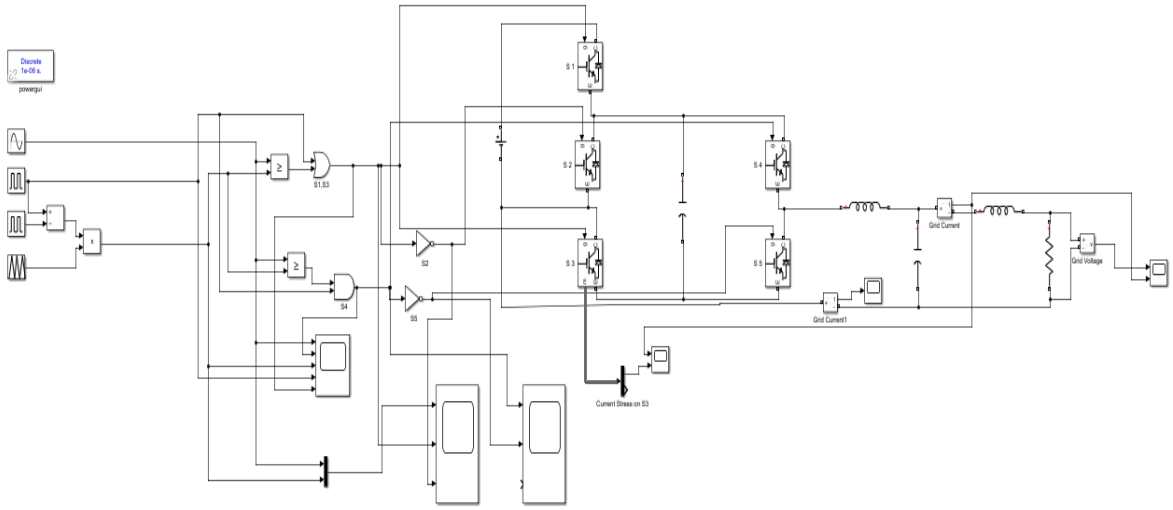


Figure 21 - Unipolar SPWM Virtual DC Bus inverter model in MATLAB/Simulink

Analysing the Figure 22 reveals that during the positive half cycle of the sinusoidal reference frequency signal the switches S_1 and S_3 remains turned ON while S_2 remains turned OFF. Meanwhile, S_4 and S_5 switches are complementarily at the carrier frequency. The PWM signal allows the output grid voltage alternates between 0 V and +380 V. During negative half cycle, the switching frequency changes to 0 to -1, with S_5 is turned on continuously and S4 is turned off. While S_1 and S_3 switches are compared with carrier frequency and S_2 operates at their complementarily switch. This allows the output alternates between 0 to -380 V.

The simulated results demonstrate that the implementation of the Virtual DC Bus aligns with the principle describe by Gu, Y. *et al.* (2013). The output voltage waveform has

three voltage levels, +380 V, 0 V and -380 V. The positive voltage levels are generated by the real bus and negative voltage level produced by the virtual bus. The virtual bus is charged during the zero-voltage level by the real bus.

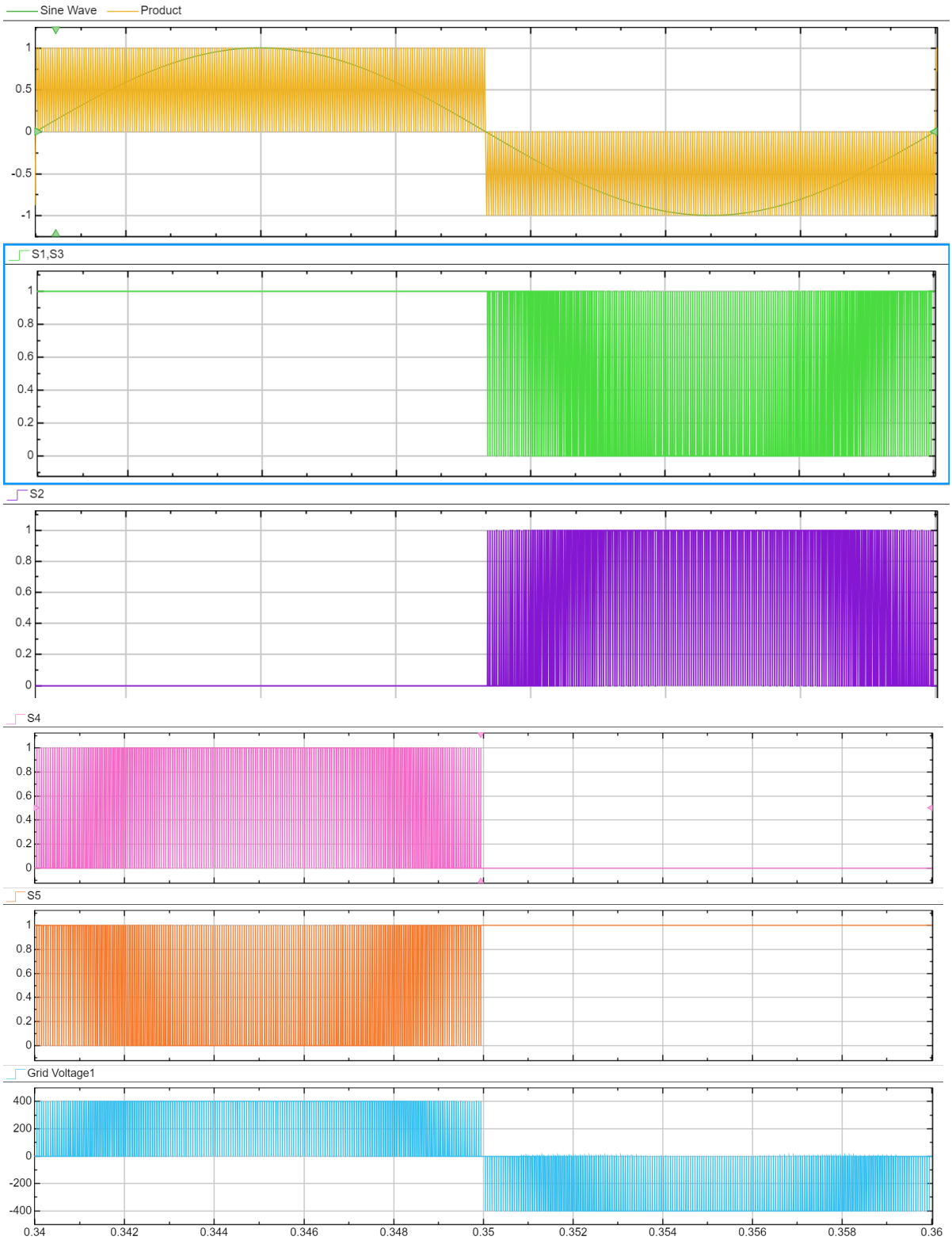


Figure 22 - Unipolar SPWM reference waveform and generated gate signals for S1, S2, S3, S4, S5 and output voltage.

The PWM output voltage shown in Figure 22 was filtered through LC filter to remove the high frequency components. The filtered results in Figure 23 shows both grid voltage and grid current achieving near sinusoidal waveforms, revealing flourish DC to AC conversion and high power factor. Small fluctuations are noticeable around the grid voltage and current, this is caused by the switching harmonics and attenuation in LC filter. The grid voltage varies between + 380 V to -380 V while grid current reaches between +38 A to -38 A, both voltage and current remain phase.

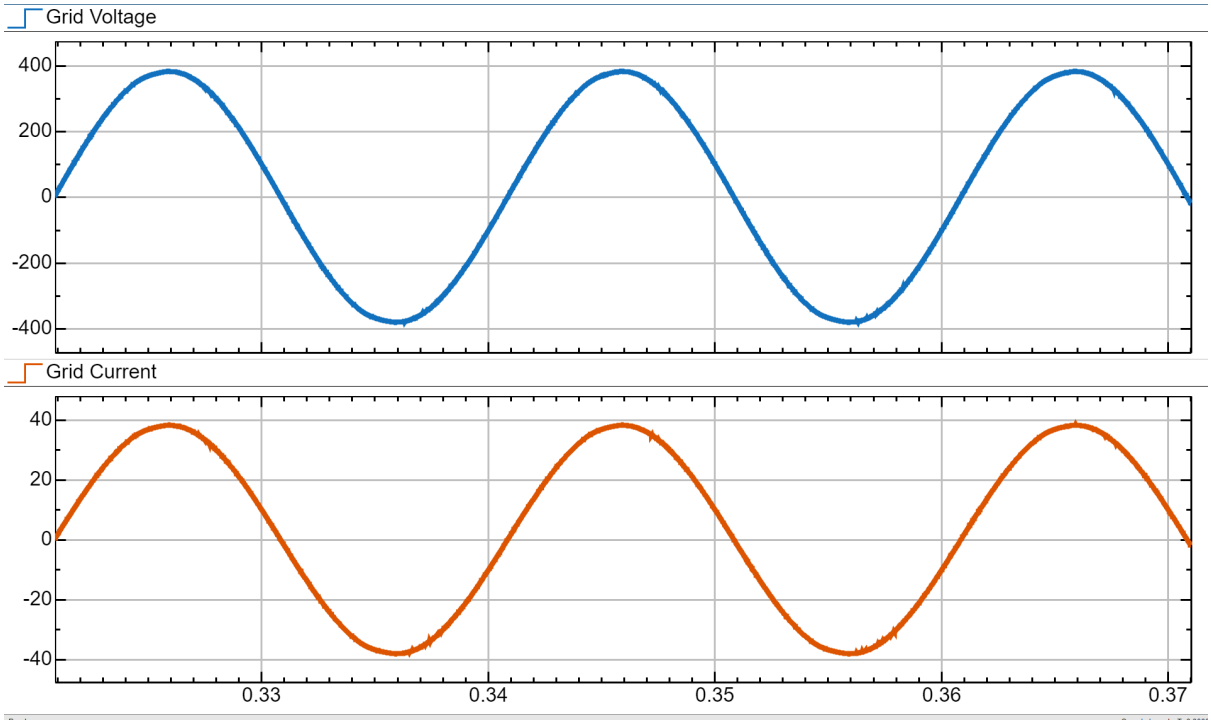


Figure 23 - Grid voltage and grid current of the unipolar SPWM Virtual DC Bus inverter

The Figure 24 shows the current stress on the switch S_3. There large current spikes occur during the negative grid half cycle, the stress during the positive half cycle is much smaller. When the real bus charges the virtual bus, S_3 experiences the highest current stress because it transmits both grid current and current to charge the capacitor.

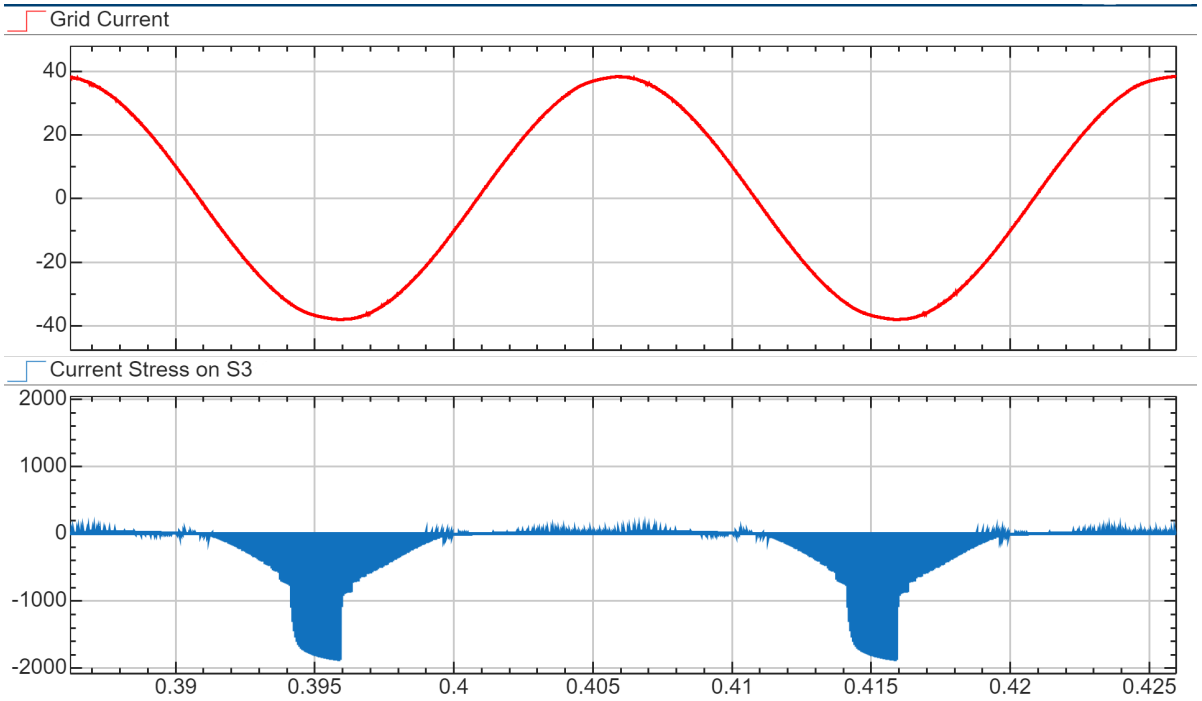


Figure 24 - current stress on switch S3 of unipolar SPWM.

6.2 Double Frequency Virtual DC Bus

Figure 25 shows the simulated switching states and output voltage before the filter was used of the Double Frequency SPWM Virtual DC Bus inverter developed in MATLAB/Simulink. The modulation technique and circuit configuration is based on Gu, Y. *et al.* (2013), where the two inverse sinusoidal signal is compared with the high frequency triangular waveform to the PWM signals.

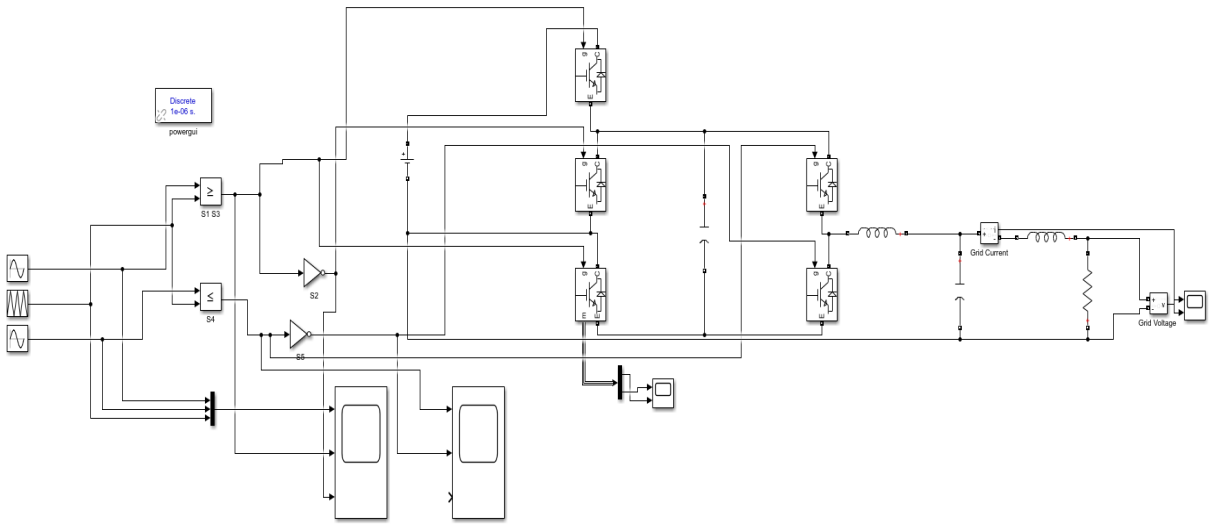


Figure 25 - Double-frequency SPWM Virtual DC Bus inverter model in MATLAB/Simulink

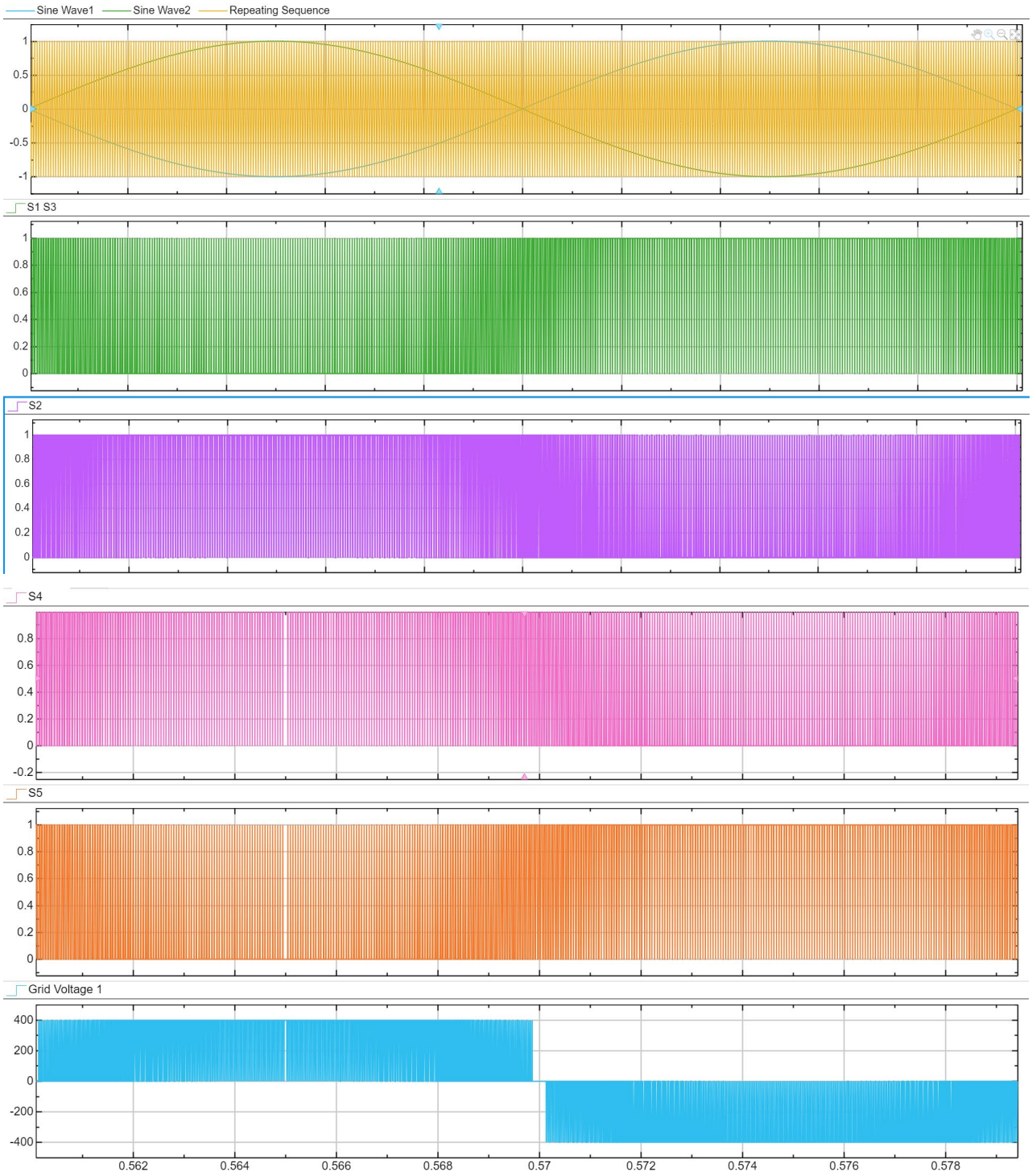


Figure 26 – Double Frequency SPWM reference waveform and generated gate signals for S1, S2, S3, S4, S5 and output voltage.

The generated switching signal are presented in Figure 26 and output voltage is shown in Figure 26. Disparate, the unipolar approach, the output voltage shifts more frequency, producing more compact PWM pattern. The output voltage is the same as the Unipolar altering between +380 V, 0 V and – 380 V.

When compared to the unipolar SPWM, the waveform is slightly smoother because switching frequency by double frequency SPWM technique. Although small fluctuation is still visible near the voltage and current peaks because of the switching harmonics and non-ideal filter, as the filtering requirement weren't provided by Gu, Y. *et al.* (2013) for double frequency SPWM. The same filter used the unipolar was utilised in this modulation technique.

The current stress on switch S_3 is shown in Figure 28. Similar to the unipolar SPWM modulation, there is highest current stress during negative half cycle when the virtual bus is being charged through S_3. Nevertheless, the effective switching frequency distributes the switching state more evenly when compared to the unipolar SPWM, there is reduction output ripple and improved harmonics. Therefore, the simulation results align with the operation principle highlighted by Gu, Y. *et al.* (2013).

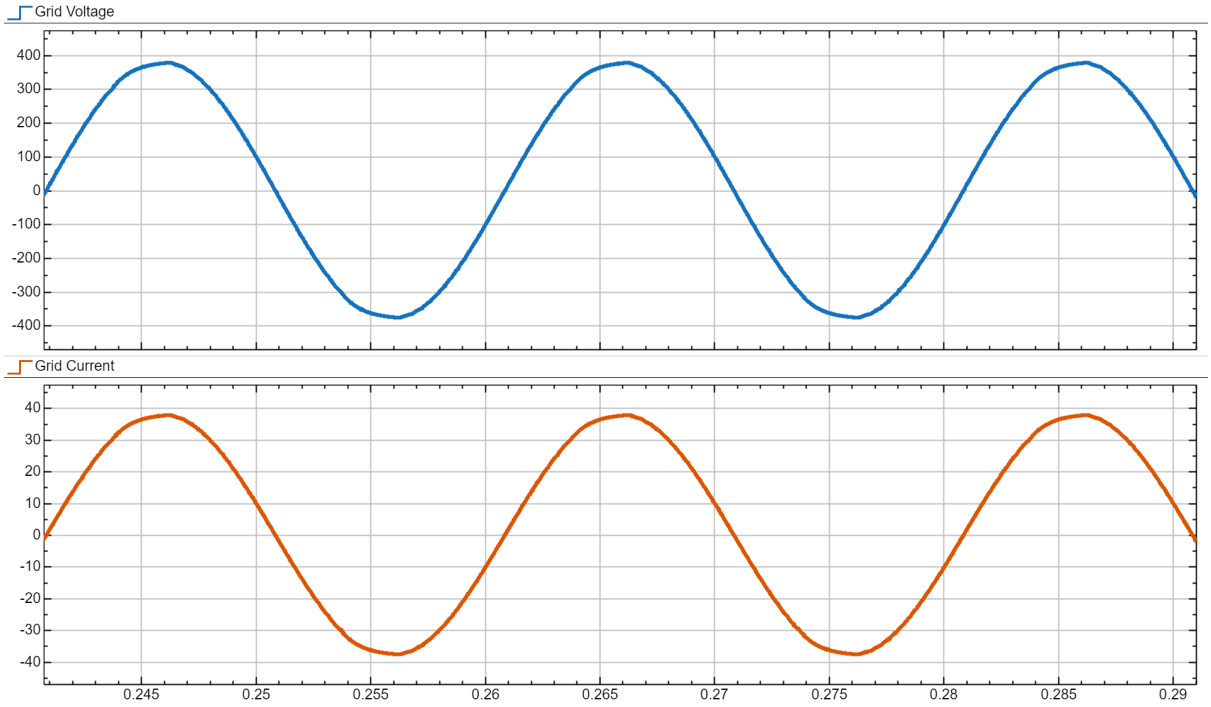


Figure 27 - Grid voltage and grid current of the double frequency SPWM Virtual DC Bus inverter

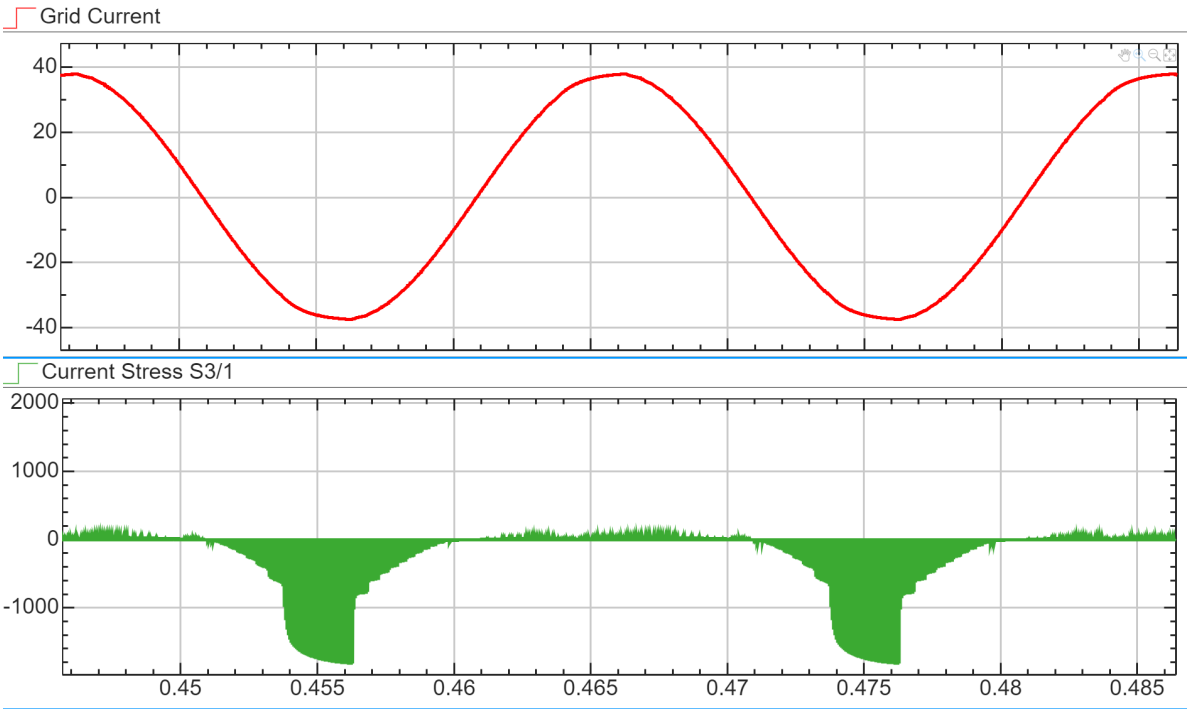


Figure 28 - current stress on switch S3 of double frequency SPWM.

As mentioned by Gu, Y. *et al.* (2013), the core advantage of double frequency SPWM is the reduction of the output current ripple, this is evident in results as the output of the unipolar has more ripple compared with double frequency modulation. Although the figure 23 and figure 27 may appear similar but internal they are different. This allows smaller filter LC filter to installed, while producing good output quality.

While as, the key advantage of unipolar is the ability to recharge the virtual bus during every switching cycle, allows better at maintaining capacitor voltage and reduces the current stress overall. Moreover, the unipolar is less complex to implement and good output waveform quality and lower switching losses than other modulation strategy.

Overall, the double frequency SPWM has superior harmonic performance and lower output ripple, whereas unipolar SPWM has simpler modulation strategy with more effective virtual bus capacitor charging.

Chapter 7 Conclusion

7.1 Summary

In conclusion, this progress report dives deeply into the transformerless PV inverter for grid connected applications, focusing on common mode (CM) voltage, leakage current, efficiency, power losses, filter requirements and modulation techniques. The aim of this report is to understand how different transformerless inverter can reduce or eliminate leakage current while gate keeping output performance and conversion efficiency. The literature review emphasizes an understanding of the operating principles and characteristics of conventional H bridged and advanced PV inverters.

The conventional H bridge inverter was initially study on as baseline topology. Both bipolar and unipolar SPWM were thought out to highlight the effects on the inverter output and common mode behaviour. The studied shows that bipolar SPWM can provide stable common mode voltage to some degree, while unipolar SPWM generate three level output voltage, current ripple and filtering requirement can be reduced. Nevertheless, unipolar modulation in transformerless inverter can produce significant leakage current path through the parasitic capacitance. This reveals that the modulation strategies have vital roles in performance of the PV inverters.

The progress project revolves around the Virtual DC Bus topology. The reviewed literature shows to eliminate the leakage current path is connect the grid neutral to the negative terminal of the PV panel to clamp the voltage across the PV parasitic capacitance. The topology operates with both unipolar and double frequency SPWM, provides three voltage levels and conceivably reduce filtering requirements. However, there are limitation to this topology as additional capacitor charging losses, ESR losses and large capacitor is required.

The Virtual DC Bus was compared with the H6 and 5-Level Mid-Point Clamped HERIC topologies to provide broader comparisons of transformerless inverter. The comparison consists of voltage level, modulation technique, operation states, filter requirements and complexity of the inverter. The findings show that referred topologies have different advantages and disadvantages. The Virtual DC Bus provides a suppression of e leakage-current path, whereas the H6 topology maintains near stable common-mode voltage to reduce the leakage current. 5-Level Mid-Point Clamped HERIC topology provides five voltage levels

and reduces the current ripple, and main drawback is the complexity of circuit and modulation technique.

Overall, the literature this study has emphasizes a understanding of key challenges associated with transformerless PV inverters and methods steps to solve them. The comparison shows that there was trade-off between leakage current suppression, efficiency, power losses, output quality and complexity. Therefore, the literature review and analysis provide a foundation for understanding and evaluating the selected transformerless PV inverter for grid connected PV applications

7.2 Timeline

Task	1 Week	2 Week	3 Week	4 Week	5 Week	6 Week	7 Week
Reseach Proposal							
Topic Selection							
Meeting with Supervisor							
Gathering Information through Research and Technical Tutorial							
Simulation of Conventional H bridge							
Writing the Research Proposal Report							
Proofreading and Correction							
Submitting the Research Proposal							

Figure 29 – Research Proposal Timeline

Task	8 Week	9 Week	10 Week	11 Week	12 Week	13 Week
Progress Report						
Meeting with Supervisor						
Further Research						
Simulation of New Topology						
Excecuting Feedback from Research Proposal						
Final Correction and Proofreading the Progress Report						
Self Reflection						
Submitting the Progress Report						

Figure 30 – Progress Report Timeline

7.3 Future Work

7.3.1 Integrating with Realistic Condition

The progress report mainly focusses the ideal condition such as DC voltage source supply used in the MATLAB/Simulink models. In the distance future, this source can be replaced with photovoltaic PV array model to represent how the PV array realistically behaves in the real world

more accurately. Subjugating the PV power generation that varies with the environmental condition, primary on sunlight, solar irradiance and temperature.

7.3.2 Implementing Battery

Introducing batteries to the PV system will be investigated in final project 2. The benefits and limitation of integrating solar battery with the PV system, how the inverter behaviour under different direction of power flow including the PV system and charging and discharging the battery. Power management system can also be paired, to coordinate electrical energy between the PV array, battery and grid.

7.3.3 Printed Circuit Board (PCB)

Progress on to the next stage of the simulation is to design PCB that can be used in the under real conditions. PCB designs for the selected inverter topology will be development based on the industry. The PCB will include components are available in the electrical market, it includes power stage, gate driver, sensor, protection circuits and control interface. The PCB will be designed in Altium designer

Appendix A

ENGR4041 Final Year Project 1 (UG Engineering) – Self-Reflection Report

To be completed and submitted as an Appendix to the Progress Report

Student Name: Mohamed Ibrahim Iqbal

Student ID: 22128421

Project Title: Analysis and comparison of Photovoltaic (PV) Inverters for Grid Applications

Supervisor(s): Dr Saad Hasan

1. Overall Project Progress and Achievements

- Reflect on your progress in the project so far.
- What stages or milestones have you reached?
- What do you consider to be your key accomplishments in this phase?

Answer:

Overall, my progress of the project has been satisfactory. I had very limited knowledge about PV inverters and transformless inverter topologies. I felt like a baby learning to crawling, I started learning the basic of inverter and moved on to researching about PV inverter topologies and SPWM. I very like this paper "Analysis and Modelling of Transformerless Photovoltaic Inverter Systems" by Tamas Kerekes, the paper had very detailed explanation and even person with no knowledge of PV inverter will have some level knowledge when they read this paper. I have taken a lot of inspiration from that paper. My key accomplishment so far has been developing good understanding of PV inverter and becoming better writer and being comfortable with MATLAB/Simulink.

2. Engagement with Literature and Context

- Describe how you explored and evaluated existing knowledge.

- How has the literature informed your project?
- What have you learned about your topic area and its relevance to engineering?

Answer:

I have explored and evaluated research papers related to PV inverters, transformerless inverters and SPWM. The literature aided me to understand the different inverter topologies and limitations associated with transformerless PV systems, especially leakage current and efficiency. Also, I have gained understanding of the purpose of comparing different topologies and how I can use the information to develop my project.

3. Skills Development

- Reflect on the skills you have developed through the project.
- What technical, analytical, communication or project management skills have you strengthened?

Answer:

I have developed my technical skills through using MATLAB/Simulink to create and develop H bridge and Virtual DC Bus. I have improved my understanding of modulation techniques such as bipolar, unipolar and double frequency SPWM. I have gotten better at conducting research and report writing through active reading research paper and online tutorials. Also, regular meeting with the supervisor have helped me improve communication, project management and how to reflect on feedback provided.

4. Challenges and Strategies

- Identify the main challenges you have encountered.
- How did you address them?
- What strategies did you find effective in progressing the project?

Answer:

Main challenges I have encountered was getting the simulated inverter to work properly. I have made several mistakes when developing the Virtual DC Bus. I addressed the problem through help page in MATLAB/Simulink, online tutorials and discussing the issues with my

supervisor. The feedback my supervisor, really helped me to understand the simulation better and make meaningful progress.

5. Future Planning

- Outline your next steps for the second half of the project.
- What are your priorities and timeline?
- How will you ensure the project stays on track?

Answer:

For next step of the second half of the project, I am planning to develop and test my selected inverter topology in MATLAB/ Simulink implement all the topologies that are discussed rather than just one. I will focus on refining my report and simulation. I also planning to simulated inverter in realistic condition to see how it behaves in the real world.

6. Meeting Minutes/Summary

Provide meeting minutes/summary.

No Meeting - Week 1

Date: 03/07/2026 (Thursday)
Duration: Approximately 0 minutes

I selected my project topic after doing research to some extent, Analysis and Comparison of Photovoltaic (PV) Inverters for Grid Applications.

Meeting 1 - Week 2

Date: 07/07/2026
Duration: Approximately 10 minutes

This was the first meeting with my supervisor; we discuss about the topic in terms of difficult level, the requirement and the initial steps. We also discussed about my current level of; I told he I have very limited knowledge in PV inverter and MATLAB/ Simulink. So, he recommended me to do some research on the PV inverter, download Simulink in MATLAB and myself familiar with it.

Meeting 2 – Week 3

Date: 14/07/2026 (Tuesday)
Duration: Approximately 20 minutes

I showed research paper that found on internet, he said the paper are not good quality. So, he showed me how to find high-quality research papers. He recommended me to use IEEE as primary source to find research papers. He gave overview of PV inverter what are transformer and transformerless PV inverter. He told me to work on developing a full bridge inverter in MATLAB/Simulink

Meeting 3 - Week 4

Date: 21/07/2026 (Tuesday)

Duration: Approximately 30 minutes

My supervisor reviewed my progress, and I made some adjustments to H bridge inverter based on his feedback. I added SPWM to my design using sine wave generator and sawtooth as input to relational operator, its output feed to the H bridge. I also create the format of the report, including heading, title page and page numbers. I have researched the internal resistance and transformerless PV inverters.

Meeting 4 - Week 5

Date: 28/07/2026 (Tuesday)

Duration: Approximately 25 minutes

I showed my simulation design to my supervisor; he identified issues with my design. He recommended an online tutorial to improve my understanding of SPWM. I went through the tutorial and fixed my simulation to implement proper SPWM. I have replaced MOSFETs with IGBTs because the IGBTs are better at handing in power. We discussed about the transformerless, report structure, formatting, title page and report writing. I also reviewed unipolar and bipolar SPWM the theory behind it.

Meeting 5 - Week 6

Date: 4/08/2026 (Tuesday)

Duration: Approximately 15 minutes

I continued working on the MATLAB/Simulink simulation and I gave overview about theory behind PWM to my supervisor. He was quite happy with my explanation. He recommended me to filling report section and developing the written content. The main focus in this meeting was improving my understanding of the simulation and checking the results were aligning with the theory.

Meeting 6 - Week 7

Date: 11/08/2026 (Tuesday)

Duration: Approximately 30 minutes

I showed my report and simulation to my supervisor; he identified some formatting issue. Advised was given to use justified alignment, to make the figures in centred position and should have even distribution tables and maintain consistent line spacing and structure. I also told use light theme for simulation results to like more professional. I made changes to the report from his feedback.

Meeting 7 - Week 8

Date: 18/08/2026 (Tuesday)

Duration: Approximately 15 minutes

The supervisor gave an overview of how the proposed report should be formatted. I was given the task of choosing a suitable inverter topology for the simulating topology in MATLAB/Simulink. This gave me a clear outline to focus on for the rest of the week.

Meeting 8 - Week 9

Date: 25/08/2026 (Tuesday)

Duration: Approximately 25 minutes

I have chosen Virtual DC Bus as my topology and showed my MATLAB/Simulink of simulation of it. There were several mistakes that were identified and discussed with the supervisor. One issue was using the Sawtooth instead of using repeating sequence to achieve triangular waveform. I corrected the issue and continued developing my simulation.

Meeting 9 - Week 10

Date: 1/09/2026 (Tuesday)

Duration: Approximately 25 minutes

We discussed the selected topology and reviewed the simulation. I ask the supervisor how to check the stress on the S_3 in the Virtual DC Bus. He advised using demux element to the measurement port to find the voltage and current stress. He also advised that I can reuse the research report to make my progress report, also he suggest changing the report accordingly the rubric of the progress report.

Meeting 10 - Week 11

Date: 8/09/2026 (Tuesday)

Duration: Approximately 40 minutes

We discuss about the comparison topologies for this progress project. I initially selected H6 and A Five-Level Common Ground Inverter with Reduced Switch Count for Transformerless Grid-Tied PV Applications. However, I wasn't sure if whether the five level inverter topology was appropriate for the current project. I have asked my supervisor for option on it, he advised that the topology was advanced and would more suitable only for postgraduate research. So, I considered the HERIC as alternative comparison topology and discussed it with my supervisor, he agreed and said it is good topology. This really helped me to niche down comparison topologies.

Meeting 11 - Week 12

Date: 15/09/2026 (Tuesday)

Duration: Approximately 15 minutes

My supervisor told me to work on the HERIC and H6 the comparison topologies in MATLAB/Simulink, then I include in the report. He advised working on the report first and then work on the comparison simulation only if there any time left. Therefore, I planned to focus on completing the report before working the simulation work.

Meeting 12 - Week 13

Date: 22/09/2026 (Tuesday)

Duration: Approximately 30 minutes

I shared my final draft with my supervisor and asked for feedback on the report and simulation. My supervisor gave advised improving my overall flow and readability, strengthen the conclusion and write a perfect abstract, the reader should know what topic I am talking about. I told this feedback to make final improvement to the report.

Declaration

I confirm that this self-reflection is my original work. No unauthorised assistance, including the use of generative AI, has been used.

Signature: Mohamed Ibrahim Iqbal Date: 25/09/2026

Appendix B

Final Project 1 - MATLAB Simulink - Simulation

References

1. Araújo, S.V., Zacharias, P. and Mallwitz, R. (2010) "Highly efficient single-phase transformerless inverters for grid-connected photovoltaic systems," *IEEE Transactions on Industrial Electronics*, 57(9), pp. 3118–3128. Available at: https://doi.org/10.1109/TIE.2009.2037654.
2. Barzegarkhoo, R. *et al.* (2023) "New Family of Dual-Mode Active Neutral Point-Clamped Five-Level Converters," *IEEE Transactions on Power Electronics*, 38(10), pp. 12236–12253. Available at: https://doi.org/10.1109/TPEL.2023.3283694.
3. Carrasco, J.M. *et al.* (2006) "Power-electronic systems for the grid integration of renewable energy sources: A survey," *IEEE Transactions on Industrial Electronics*, pp. 1002–1016. Available at: https://doi.org/10.1109/TIE.2006.878356.
4. Cecati, C., Ciancetta, F. and Siano, P. (2010) "A multilevel inverter for photovoltaic systems with fuzzy logic control," *IEEE Transactions on Industrial Electronics*, 57(12), pp. 4115–4125. Available at: https://doi.org/10.1109/TIE.2010.2044119.
5. Cedieu, S. *et al.* (2022) "Four-Switch Five-Level Common-Ground Transformerless Inverter," *2022 14th Seminar on Power Electronics and Control, SEPOC 2022*. Institute of Electrical and Electronics Engineers Inc. Available at: https://doi.org/10.1109/SEPOC54972.2022.9976457.
6. *CMCE 2010 : 2010 International Conference on Computer, Mechatronics, Control and Electronic Engineering : August 24-26, 2010, Changchun, China* (2010). IEEE.
7. Freddy, T.K.S. *et al.* (2014) "Comparison and analysis of single-phase transformerless grid-connected PV inverters," *IEEE Transactions on Power Electronics*, 29(10), pp. 5358–5369. Available at: https://doi.org/10.1109/TPEL.2013.2294953.
8. Gu, B. *et al.* (2013) "High reliability and efficiency single-phase transformerless inverter for grid-connected photovoltaic systems," *IEEE Transactions on Power Electronics*, 28(5), pp. 2235–2245. Available at: https://doi.org/10.1109/TPEL.2012.2214237.

9. Gu, Y. *et al.* (2013) "Transformerless inverter with virtual DC bus concept for cost-effective grid-connected PV power systems," *IEEE Transactions on Power Electronics*, 28(2), pp. 793–805. Available at: https://doi.org/10.1109/TPEL.2012.2203612.
10. Guan, Z. and Tian, F. (2020) "Design and Implementation of Energy Storage Photovoltaic Grid-Connected Power Generation System," *2020 IEEE 3rd International Conference on Automation, Electronics and Electrical Engineering, AUTEEE 2020*. Institute of Electrical and Electronics Engineers Inc., pp. 61–66. Available at: https://doi.org/10.1109/AUTEEE50969.2020.9315693.
11. IEEE Staff, . (2011) *2011 International Conference on Electrical and Control Engineering*. IEEE.
12. Ji, B., Wang, J. and Zhao, J. (2013) "High-efficiency single-phase transformerless PV H6 inverter with hybrid modulation method," *IEEE Transactions on Industrial Electronics*, 60(5), pp. 2104–2115. Available at: https://doi.org/10.1109/TIE.2012.2225391.
13. Kerekes, T. (2009) *Analysis and Modeling of Transformerless Photovoltaic Inverter Systems*.
14. Kerekes, T. *et al.* (2011) "A New high-efficiency single-phase transformerless PV inverter topology," *IEEE Transactions on Industrial Electronics*, 58(1), pp. 184–191. Available at: https://doi.org/10.1109/TIE.2009.2024092.
15. Khadem, S.M., Yeh, H.G. and Vosoughi, N. (2026) "A Novel Nine-Level Grid-Tied Common-Ground Inverter with Reactive Power Control, Using One-Cycle Method," *Conference Proceedings - IEEE Applied Power Electronics Conference and Exposition - APEC*. Institute of Electrical and Electronics Engineers Inc., pp. 2699–2706. Available at: https://doi.org/10.1109/APEC51134.2026.11516906.
16. Khan, M.N.H. *et al.* (2020) "Transformerless Inverter Topologies for Single-Phase Photovoltaic Systems: A Comparative Review," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 8(1), pp. 805–835. Available at: https://doi.org/10.1109/JESTPE.2019.2908672.
17. Kouro, S. *et al.* (2015) "Grid-connected photovoltaic systems: An overview of recent research and emerging PV converter technology," *IEEE Industrial Electronics Magazine*, 9(1), pp. 47–61. Available at: https://doi.org/10.1109/MIE.2014.2376976.
18. Li, L., Zhang, L. and Zhao, Y. (2021) "Dynamic Equivalent Modeling of Photovoltaic Grid-connected Power Generation System," *Proceedings - 2021 6th Asia Conference on Power and Electrical Engineering, ACPEE 2021*. Institute of Electrical and

Electronics Engineers Inc., pp. 1583–1587. Available at: https://doi.org/10.1109/ACPEE51499.2021.9436872.

19. López, Ó. *et al.* (2010) "Eliminating ground current in a transformerless photovoltaic application," *IEEE Transactions on Energy Conversion*, 25(1), pp. 140–147. Available at: https://doi.org/10.1109/TEC.2009.2037810.
20. Madhavan, M. *et al.* (2025) "A Study on the Performance Evaluation of F-Type Multilevel Inverters Employing Phase Disposition Carrier PWM Schemes," *IEEE Access*, 13, pp. 48286–48305. Available at: https://doi.org/10.1109/ACCESS.2025.3549589.
21. Meneses, D. *et al.* (2013) "Review and comparison of step-up transformerless topologies for photovoltaic AC-module application," *IEEE Transactions on Power Electronics*, 28(6), pp. 2649–2663. Available at: https://doi.org/10.1109/TPEL.2012.2227820.
22. Romero-Cadaval, E. *et al.* (2013) "Grid-connected photovoltaic generation plants: Components and operation," *IEEE Industrial Electronics Magazine*, 7(3), pp. 6–20. Available at: https://doi.org/10.1109/MIE.2013.2264540.
23. Siwakoti, Y.P. and Blaabjerg, F. (2018) "Common-ground-type transformerless inverters for single-phase solar photovoltaic systems," *IEEE Transactions on Industrial Electronics*, 65(3), pp. 2100–2111. Available at: https://doi.org/10.1109/TIE.2017.2740821.
24. Syasegov, Y.Y. *et al.* (2022) "A 5-Level Mid-Point Clamped HERIC Inverter," *2022 IEEE 13th International Symposium on Power Electronics for Distributed Generation Systems, PEDG 2022*. Institute of Electrical and Electronics Engineers Inc. Available at: https://doi.org/10.1109/PEDG54999.2022.9923145.
25. Wang, E., Ma, X. and Liu, H. (2025) "Analysis of Grid-Connected Photovoltaic Systems Based on Hybrid Energy Storage," *2025 IEEE 7th International Conference on Power, Intelligent Computing and Systems, ICPICS 2025*. Institute of Electrical and Electronics Engineers Inc., pp. 364–368. Available at: https://doi.org/10.1109/ICPICS66386.2025.11347306.
26. Wu, L., Zhao, Z. and Liu, J. (2007) "A single-stage three-phase grid-connected photovoltaic system with modified MPPT method and reactive power compensation," *IEEE Transactions on Energy Conversion*, 22(4), pp. 881–886. Available at: https://doi.org/10.1109/TEC.2007.895461.
27. Xiao, H. *et al.* (2011) "An optimized transformerless photovoltaic grid-connected inverter," *IEEE Transactions on Industrial Electronics*, 58(5), pp. 1887–1895. Available at: https://doi.org/10.1109/TIE.2010.2054056.

28. Yang, B. *et al.* (2010) "Design and analysis of a grid-connected photovoltaic power system," *IEEE Transactions on Power Electronics*, 25(4), pp. 992–1000. Available at: https://doi.org/10.1109/TPEL.2009.2036432.
29. Yang, B. *et al.* (2012) "Improved transformerless inverter with common-mode leakage current elimination for a photovoltaic grid-connected power system," *IEEE Transactions on Power Electronics*, 27(2), pp. 752–762. Available at: https://doi.org/10.1109/TPEL.2011.2160359.
30. Yang, Y., Blaabjerg, F. and Wang, H. (2014) "Low-voltage ride-through of single-phase transformerless photovoltaic inverters," *IEEE Transactions on Industry Applications*, 50(3), pp. 1942–1952. Available at: https://doi.org/10.1109/TIA.2013.2282966.
31. Yu, J. *et al.* (2023) "Study on Small Disturbance Stability of Photovoltaic Grid-Connected Power Generation System," *2023 5th Asia Energy and Electrical Engineering Symposium, AEEES 2023*. Institute of Electrical and Electronics Engineers Inc., pp. 1429–1433. Available at: https://doi.org/10.1109/AEEES56888.2023.10114185.
32. Zhan, S. *et al.* (2024) "Application of Key Measurement and Control Technologies for Distributed Photovoltaic Grid Connected Systems," *Proceedings - 2024 2nd International Conference on Mechatronics, IoT and Industrial Informatics, ICMIII 2024*. Institute of Electrical and Electronics Engineers Inc., pp. 926–931. Available at: https://doi.org/10.1109/ICMIII62623.2024.00178.
33. Zhang, L. *et al.* (2014) "H6 transformerless full-bridge PV grid-tied inverters," *IEEE Transactions on Power Electronics*, 29(3), pp. 1229–1238. Available at: https://doi.org/10.1109/TPEL.2013.2260178.
34. U.S. Energy Information Administration 2024, *Photovoltaics and Electricity*, *Eia.gov*, 24 May, U.S. Energy Information Administration, viewed 25 July 2026, <https://www.eia.gov/energyexplained/solar/photovoltaics-and-electricity.php>.
35. *Electricity production by source* (2026) *Our World in Data*. Available at: https://ourworldindata.org/grapher/electricity-production-by-source (Accessed: 01 August 2026).
36. Freddy, T.K.S. *et al.* (2015) "Modulation techniques to reduce leakage current in three-phase transformerless H7 photovoltaic inverter," *IEEE Transactions on Industrial Electronics*, 62(1), pp. 322–331. Available at: https://doi.org/10.1109/TIE.2014.2327585.
37. Freddy, T.K.S. *et al.* (2017) "Modulation Technique for Single-Phase Transformerless Photovoltaic Inverters with Reactive Power Capability," *IEEE Transactions on*

Industrial Electronics, 64(9), pp. 6989–6999. Available at: https://doi.org/10.1109/TIE.2017.2686366.

38. Ozkan, Z. and Hava, A.M. (2011) 'Leakage current analysis of grid connected transformerless solar inverters with zero vector isolation', *2011 IEEE Energy Conversion Congress and Exposition*, pp. 2460–2466. doi:10.1109/ecce.2011.6064095.
39. IoTDunia (2025) *What is Smart Grid Technology? beginner's guide 2025*, *What is smart grid technology*. Available at: https://iotdunia.com/what-is-smart-grid-technology/ (Accessed: 02 August 2026).
40. Sopian, K.B. *et al.* (2004) *Photovoltaic Panel - An Overview | Sciencedirect topics*, *Photovoltaic Panel*. Available at: https://www.sciencedirect.com/topics/engineering/photovoltaic-panel (Accessed: 09 August 2026).
41. Donev, J. et al. (2024) Photovoltaic cell, Energy Education. Available at: https://energyeducation.ca/encyclopedia/Photovoltaic_cell (Accessed: 09 August 2026).